

1 Irrigation strategies under contrasting water-storage and rainfall

2 conditions in the North China Plain

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9 Cover letter

10 Dear Editor,

11 Please consider the research article “Irrigation strategies under contrasting water-storage and rainfall conditions in
12 the North China Plain” for publication in Agricultural Water Management.

13 Reducing irrigation while sustaining grain production requires balancing water consumption and the distribution of
14 production losses. Using a process-based crop model, the study compares uniform and spatially targeted irrigation
15 under equal water budgets and tests whether satellite water-storage information improves rainfall-based irrigation
16 decisions.

17 At the same 50% irrigation reduction, targeting retained 4.87% more simulated grain than uniform cuts, but
18 increased evapotranspiration and reduced grain production in some locations. Adding regional storage information
19 did not change irrigation quotas or outcomes relative to matched rainfall-only strategies within the tested design.

20 These model-based findings support irrigation strategies that combine production goals with limits on consumptive
21 use and safeguards for local production. Their focus on irrigation allocation and crop water use makes the study
22 well suited to Agricultural Water Management.

23 Thank you for considering this manuscript.

24 Sincerely,

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29 **Highlights**

- 30 • Targeting retained 0.732 t ha⁻¹ yr⁻¹ more dry grain than uniform 50% cuts.
- 31 • The same irrigation budget produced 20.76 mm yr⁻¹ more simulated ET.
- 32 • At 50% cuts, mean grain was below uniform allocation on 37.60% of the area.
- 33 • Over nine GRACE years, grain retention was 96.27% (95% target) and 99.63% (98%).
- 34 • Matched rainfall-only controls differed in grain production by 0% at both targets.

35 **Manuscript**

36 **Abstract**

37 Efficient irrigation allocation is essential for sustaining grain production while limiting water consumption in the
38 North China Plain wheat–maize rotation. How spatial targeting redistributes production losses, and whether
39 regional water-storage information improves rainfall-based irrigation decisions, remain unresolved. Continuous
40 process-based crop simulations compared uniform and targeted allocations under equal irrigation budgets and
41 evaluated annual storage–rainfall strategies against matched rainfall-only controls during 2014–2025. Growth
42 parameters were fitted to multisite observations; water-use parameters were calibrated against Wuqiao treatments in
43 2016–2018 and retrospectively tested in 2019. At a 50% irrigation reduction, targeting retained 0.73 t ha⁻¹ yr⁻¹
44 more simulated combined dry grain than uniform cuts across 9.87 million rotation hectares, equivalent to 7.22
45 million t yr⁻¹ or 4.87% of uniform production. Mean production was lower than under uniform cuts on 37.60% of
46 mapped area, while crop-plus-fallow evapotranspiration increased by 20.76 mm yr⁻¹ (2.05 km³ yr⁻¹). Across nine
47 years with eligible Gravity Recovery and Climate Experiment (GRACE) observations, strategies selected at 95%
48 and 98% grain targets retained 96.27% and 99.63% of conventional production, respectively. The 95% strategy
49 reduced irrigation by 130.72 mm and evapotranspiration by 36.84 mm per rotation. Matched storage–rainfall and

50 rainfall-only strategies selected identical quotas and produced identical outcomes. These conditional results support
51 joint production and consumptive-use constraints and local production safeguards, while showing that additional
52 storage information provides no decision benefit within the tested quota design.

53 **Keywords**

54 winter wheat–summer maize rotation; irrigation management; crop modeling; GRACE; water availability;
55 consumptive water use

56 **1. Introduction**

57 Irrigation management in the North China Plain connects grain production with groundwater availability across the
58 winter wheat–summer maize rotation (Yang et al., 2022). Wheat grows through a relatively dry season, whereas
59 maize receives more summer rainfall; soil-water reserves link their responses to water supply (Yang et al., 2024).
60 Long-term well-watered observations show that rising grain yield has accompanied increasing actual
61 evapotranspiration (ET), despite comparatively stable reference ET (Zhang et al., 2011). Recent aquifer recovery
62 under water transfers, abstraction management and changing climate further emphasizes the connection between
63 agricultural production and regional water-resource objectives (Long et al., 2025; Xu et al., 2026). Irrigation
64 decisions consequently involve the water delivered to fields, the grain it supports and the water consumed over the
65 complete rotation.

66 The production return from that delivery depends on rainfall and preceding crop management. A sixteen-year
67 experiment in 2003–2018 compared seven wheat–maize irrigation schedules and found diminishing marginal
68 production gains as irrigation increased (Yang et al., 2022). A four-year Wuqiao experiment showed that
69 precipitation conditions influenced the preferred wheat schedule and grain production in the subsequent maize crop
70 (Yang et al., 2024). Water applied to wheat can support its immediate growth and leave a different soil-water
71 reserve for maize. The annual response to an irrigation reduction therefore depends on the sequence of crop
72 demand, rainfall and storage changes across both seasons (Yang et al., 2024; Yang et al., 2022).

73 These temporal dependencies occur within a spatially heterogeneous region. Irrigation requirements vary across the
74 North China Plain and among hydrological years in the broader Huang–Huai–Hai Plain (Ren et al., 2022; Wu et al.,
75 2019). Regional crop-model optimization and irrigation-allocation studies demonstrate how climate, soil and crop
76 responses can inform management choices (Jiang et al., 2016; Xiao et al., 2024). Under a limited regional supply,
77 spatial allocation depends on the additional grain obtained from an additional unit of irrigation. An equal-budget

78 comparison with uniform reductions can quantify the production benefit of retaining water in more responsive
79 systems. That benefit also has a water-accounting and distributional dimension: delivery, ET, return flows and
80 storage change represent different boundaries, and a regional grain gain can coexist with losses at particular
81 locations (Grafton et al., 2018; Perry et al., 2009).

82 Quantifying these trade-offs requires crop responses that connect canopy development, grain formation and soil-
83 water depletion through successive seasons. Applications of APSIM and RZ-SHAW in the North China Plain
84 provide process-based assessments of crop water productivity, field water balance and ET (Chen et al., 2010; Fang
85 et al., 2014). Transpiration formulation is a major source of variation in simulated water use among wheat models
86 (Cammarano et al., 2016). Yield agreement alone therefore provides incomplete evidence about the water cost of
87 production or the production response to additional irrigation. Phenology, canopy, biomass, grain and ET
88 observations constrain complementary processes, while documented irrigation treatments identify their combined
89 response to water supply. Continuous simulation preserves the soil-water carry-over required to compare complete
90 management histories.

91 Spatial allocation determines where a regional irrigation budget is delivered; antecedent environmental information
92 can also determine how large that budget should be in a particular year. Recent rainfall describes water input,
93 whereas GRACE total water storage integrates several terrestrial stores over a regional footprint (Save et al., 2016;
94 Scanlon et al., 2016). The storage indicator can thus describe conditions beyond local rainfall, but additional
95 environmental information has management value only when it changes the selected action or its grain–water
96 outcome. Storage–rainfall strategies require comparison with rainfall-only strategies under common selection
97 criteria and monitoring availability. This comparison separates the physical relevance of a hydrological indicator
98 from its incremental value for the available irrigation choices.

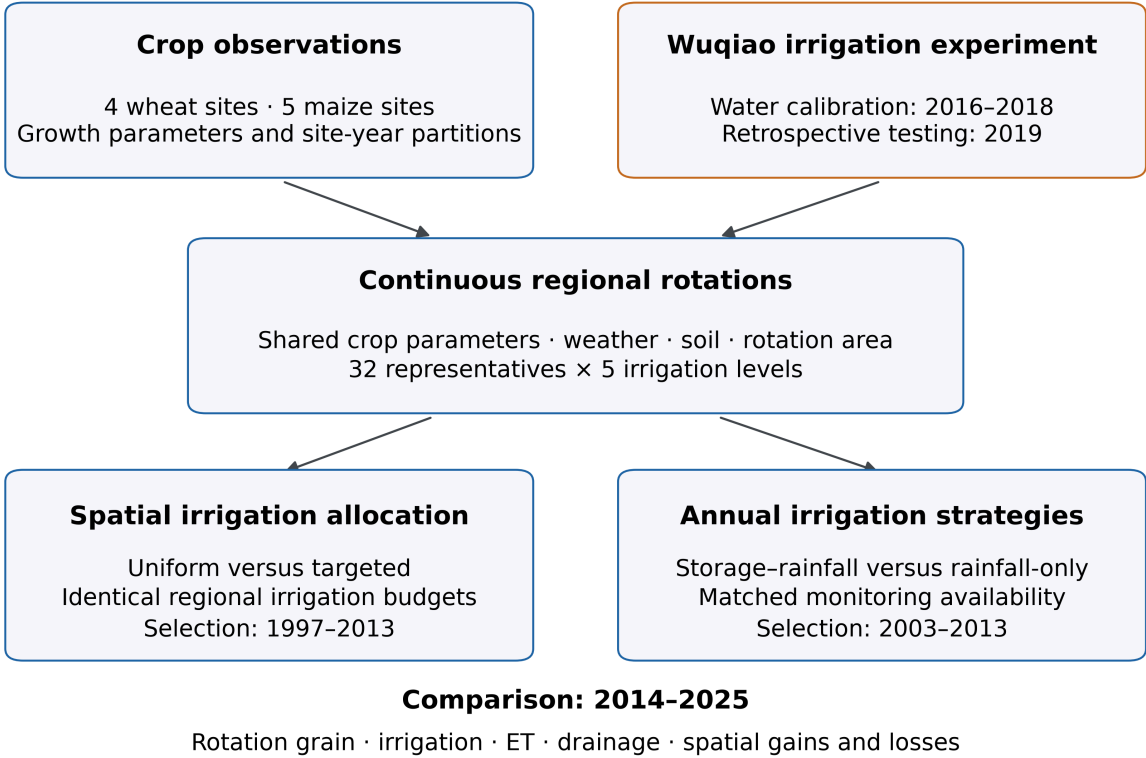
99 Spatial irrigation allocation and antecedent rainfall–storage strategies were compared for the North China Plain
100 wheat–maize rotation using Open Crop Model. Multisite observations supplied crop-growth parameter estimates,
101 and documented Wuqiao treatments in 2016–2018 constrained water-response coefficients. The 2019 field
102 observations provided retrospective testing, while station comparisons characterized transfer under their available
103 management information. Continuous regional simulations compared uniform and targeted allocations under
104 identical irrigation budgets and annual storage–rainfall strategies against matched rainfall-only controls. Production
105 retention, consumptive ET, profile drainage and the spatial and annual distribution of grain changes define the
106 management trade-offs.

107 **2. Materials and methods**

108 **2.1. Overall workflow**

109 The framework connects crop-response estimation with spatial allocation and annual irrigation-quota selection
110 (Figure 1). Multisite crop observations support growth-parameter estimates, while documented Wuqiao treatments
111 in 2016–2018 constrain eight water coefficients per crop. Wuqiao 2019 provides retrospective testing; station
112 observations characterize transfer under the recorded management and initialization. Observation coverage and the
113 distinct estimation and evaluation partitions are summarized in Figure 3. Daily weather, mapped rotation area and
114 layered soil profiles define 32 regional representatives.

115 The same crop and water parameters drive continuous crop and fallow histories under five irrigation levels. Spatial
116 allocations selected from 1997–2013 responses share identical regional delivery budgets with uniform reductions.
117 Annual storage–rainfall and rainfall-only strategies are selected from matched 2003–2013 responses and compared
118 with common monitoring availability. Grain production, field irrigation, crop-plus-fallow ET and profile drainage
119 are evaluated during 2014–2025.



120 **Figure 1. Study workflow.** Multisite observations support growth parameters, while documented Wuqiao treatments calibrate water-
121 use parameters and provide retrospective testing. Shared parameters drive continuous regional rotations. Spatial allocation
122 compares uniform and targeted irrigation under identical budgets; annual strategies compare storage–rainfall and rainfall-only

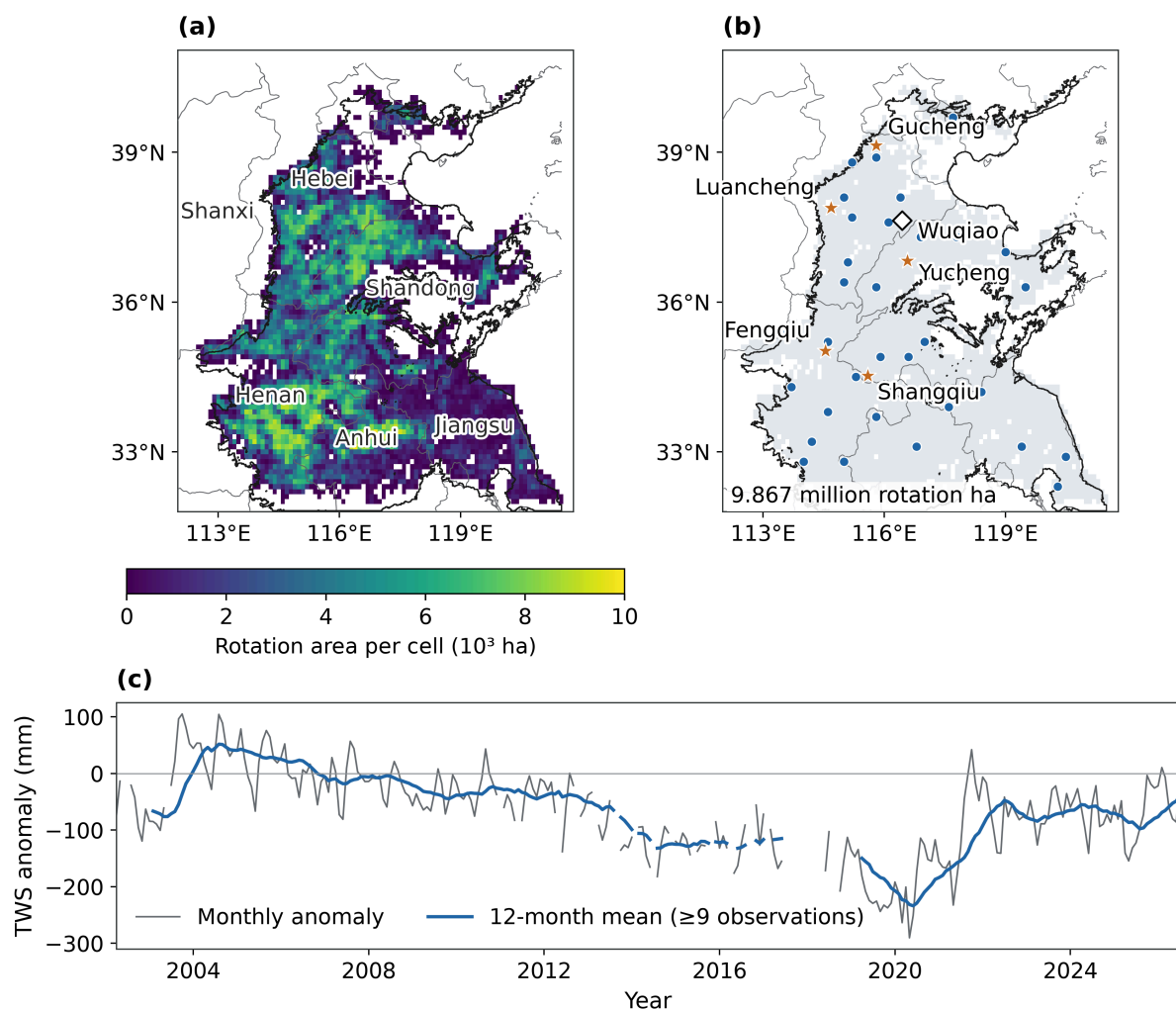
123 selection under matched monitoring availability. Outcomes cover 2014–2025.

124 **2.2. Study area**

125 The study area covers the winter-wheat–summer-maize lowlands of the North China Plain within 112–121.5°E and
126 32–41°N, including the Beijing–Tianjin–Hebei sector and the Henan–Anhui–Jiangsu lowlands (Figure 2).

127 Catchment screening, elevation ≤ 200 m and slope $\leq 2^\circ$ define a connected lowland area of 404,658 km². Terrain was
128 represented at 0.01° resolution using resampled SRTM elevation and derived slope (NASA JPL, 2013). The
129 boundary is a reproducible agricultural study-area definition; geographical and terrain-threshold sensitivities are
130 reported in Supplementary Section S3.

131 The 2020 ChinaCP winter-wheat–summer-maize class was corrected by the corresponding CPM cropland
132 percentage to reduce full-pixel area inflation (Qiu et al., 2022). The resulting rotation footprint was 9,866,733 ha
133 before soil exclusions. Three cells with wholly missing soil profiles, totaling 131 ha, were excluded, leaving
134 9,866,602 ha in 3,641 AgERA5 cells. A fixed 2020 footprint applies to all climate years, so modeled totals describe
135 climatic and management responses over a common area rather than historical production trends. The 2015–2021
136 area series, terrain-threshold alternatives and a common-extent comparison with CCD2020 quantify mapping and
137 boundary sensitivity in the supplementary material.



138 **Figure 2. Study domain, rotation area, observation sites and regional water storage. (a) Mapped 2020 wheat–maize rotation area. (b)**
 139 **Five crop-observation sites, the Wuqiao irrigation experiment and 32 regional simulation units. (c) Observed CSR monthly**
 140 **terrestrial-water-storage anomalies and a 12-calendar-month mean requiring at least nine observations. Missing months remain**
 141 **gaps. Thin grey lines show provincial boundaries; black outlines delimit the operational physiographic study domain. Maps use an**
 142 **equiarectangular projection (36°N, 117°E). Provincial boundaries: Natural Earth Admin 1 v5.1.1 (Natural Earth, n.d.).**

143 **2.3. Field observations and environmental data**

144 **2.3.1. Multisite crop observations**

145 Winter-wheat and summer-maize observations were assembled from the National Ecosystem Science Data Center
 146 and linked archives at Fengqiu, Gucheng, Luancheng, Shangqiu and Yucheng. The sources contain long-term crop
 147 monitoring and the Gucheng maize sowing-date experiment. Archived crop-season groups span 2004–2008 at
 148 Fengqiu and Luancheng, 2018–2020 at Gucheng, 2001–2007 at Shangqiu and 1998–2022 at Yucheng. Coverage is
 149 intermittent; exact crop-by-site years and dataset identifiers are listed in Tables S1 and S2.

150 The archive retains 132 crop–plot–season trajectories in 71 original calibration and 61 retrospective-evaluation
151 cases (Figure 3). Quality-controlled targets occur in 127 cases. The 7,397 scored records comprise 795 LAI
152 measurements, 543 in-season aboveground-biomass measurements, 5,675 daily ET measurements, 192 grain-yield
153 samples and 192 harvest-biomass samples (Table 1). Daily ET is available at Fengqiu and Yucheng; grain and
154 harvest-biomass samples are confined to Yucheng. Six quadrats within a crop-year characterize sampling variation
155 rather than six independent growing seasons.

156 Observations and predictions share recorded sampling dates or documented averaging windows. Stage-indexed
157 Gucheng LAI measurements use the corresponding observed stage dates as sampling coordinates; those dates do
158 not prescribe simulated development. Fengqiu ET identifies the lysimeter observation plot. Yucheng ET retains the
159 measurement definitions of its source archive. Provider grain and aboveground dry masses in g m^{-2} are multiplied
160 by 10 to obtain kg ha^{-1} .

161 Eligibility requires valid dates, compatible crop and plot identities, consistent stage sequences and documented
162 quantity units. Wheat LAI above $7 \text{ m}^2 \text{ m}^{-2}$ is excluded under the study-specific screening rule, with raw values and
163 reasons preserved. The six-replicate maize harvest vectors in Yucheng 2012 and 2017 are identical despite different
164 dates and cultivar labels. Both groups remain excluded because their source authenticity is unresolved; 12 samples
165 account for 24 grain and harvest-biomass records. Other observations and all eligible sites remain available.

166 Complete irrigation histories matched to station plots remain unconfirmed. Station growth and ET consequently
167 supply conditional transfer comparisons; no station observation enters the current water-parameter objective.
168 Twenty-six Fengqiu cases retain complete 0–1.5 m moisture profiles from the preceding calendar month: 11 in the
169 original calibration partition and 15 in retrospective evaluation. Volumetric percentages from fifteen 100-mm source
170 layers are conservatively remapped to the model layers without depth extrapolation or clipping. Their monthly
171 timing and subplot support do not establish a measured sowing-day state. Cases without an eligible profile retain
172 their archived initialization.

173 **Table 1. Eligible station-observation records by site and variable.**

Site	LAI	Biomass	Daily ET	Yield	Harvest biomass
Fengqiu	159	163	962	0	0
Gucheng	216	0	0	0	0
Luancheng	123	165	0	0	0
Shangqiu	114	148	0	0	0
Yucheng	183	67	4713	192	192

174 Counts denote individual observations, not independent experiments. Station comparisons retain inherited growth-
175 fitting history and unconfirmed complete management. Yucheng grain and harvest biomass contain six dry-matter
176 samples per crop-year; the duplicated 2012 and 2017 maize groups remain excluded.

177 **2.3.2. Field irrigation experiment**

178 The Wujiao experiment at 37.630°N, 116.444°E contains four wheat irrigation schedules, W0–W3, during the
179 2015–2019 rotation (Yang et al., 2024). Dated 75-mm events, sowing and harvest dates, nitrogen input of 158.4 kg
180 ha⁻¹ per crop and treatment-specific initial total water storage in 2 m are retained. Maize irrigation is identical
181 among treatments within each year: zero in 2016 and 2018, and 75 mm on 18 June in 2017 and 2019. Its W0–W3
182 labels therefore identify preceding wheat management. Seasonal initial water distribution among layers is assumed.
183 Field capacity, wilting point and bulk density are transferred from the published Luancheng profile, with its bottom
184 10 cm extrapolated (Sun et al., 2006). Conductivity follows a separate SoilGrids–Saxton–Rawls prior (ISRIC —
185 World Soil Information, 2020; Saxton and Rawls, 2006); the particle-density prior is 2.75 g cm⁻³.

186 Published seasonal ET equals rainfall plus irrigation plus depletion of 0–2 m storage, with runoff and drainage
187 treated as negligible (Yang et al., 2024). The model represents both outflows. The 2016 maize balance uses 499.1
188 mm of maize-season rainfall; the 599.6-mm rotation total also contains 100.5 mm during wheat. This period
189 reconciliation reproduces the four printed maize ET values within 0.9 mm of rounding, and all 32 crop–treatment–
190 season ET targets remain eligible. Simulations retain their exact archived weather forcing. Wheat crop-start storage
191 follows the pre-sowing simulation, whereas maize is independently initialized from observed treatment-specific
192 total storage. Reported wheat-straw mulch has no measured cover fraction, so the maize simulations retain the
193 exposed-soil evaporation boundary (Yang et al., 2024).

194 Harvest aboveground biomass includes grain and residues cut at ground level and oven-dried to constant mass. Its
195 dry-matter basis requires no moisture adjustment. Published grain yields are standardized to 13% moisture (Yang et
196 al., 2024). Annual 2016–2019 means and standard errors are digitized from the source Supplementary Figure S2
197 and reconciled with its eight printed four-year means. Pixel tolerances are approximately 0.043 t ha^{-1} for wheat and
198 0.056 t ha^{-1} for maize. Observed grain is multiplied by 0.87 for dry-matter fitting; model dry grain is divided by
199 0.87 for comparisons reported at 13% moisture. Annual and four-year means overlap and do not provide separate
200 independent observations.

201 The 2016–2018 harvest years contribute 12 treatment seasons per crop to the water fit, with seasonal ET, dry grain,
202 dry aboveground biomass and annual W3–W0 ET contrasts as targets. The four 2019 treatment means per crop are
203 excluded from fitting and parameter selection. Their prior inspection during model development makes this test
204 retrospective (Figure 3).

205 **2.3.3. Meteorological, soil and spatial data**

206 Daily AgERA5 v2.0 weather data for 1996–2025 supplied minimum and maximum air temperature,
207 precipitation, solar radiation, reference evapotranspiration, vapor pressure and 10-m wind speed
208 (Copernicus Climate Change Service, 2020). Reference evapotranspiration (ET_0) entered the water-
209 balance module directly.

210 Native 250 m SoilGrids v2.0 maps of texture, bulk density, organic carbon and coarse fragments were averaged to
211 0.01° , then aggregated over mapped rotation pixels within 0.1° meteorological cells, for six depth intervals between
212 0 and 2 m (ISRIC — World Soil Information, 2020). Hydraulic properties were derived using Saxton–Rawls
213 equations with bulk-density and gravel adjustments (Saxton and Rawls, 2006). Organic matter was approximated
214 using an assumed organic-matter/organic-carbon ratio of 1.724, and air-dry water content was assumed to equal half
215 the wilting-point value. Hydraulic derivation is detailed in Supplementary Section S2.

216 The 2020 ChinaCP wheat–maize class and its CPM cropland fractions defined rotation-area weights (Qiu et al.,
217 2022). SRTM elevation supported the terrain screen and slope calculation (NASA JPL, 2013). Regional aggregation
218 retained the common 0.1° meteorological grid and the study-area definition in Section 2.2.

219 **2.3.4. Terrestrial-water-storage observations**

CSR RL06.3 monthly GRACE/GRACE-FO total-water-storage anomalies were aggregated using physical cell areas and the official CSR land mask (Save, 2020; Save et al., 2016). The land-mask overlap covers 395,969 km² within the 404,658-km² study area; excluded coastal supports account for 2.15% of its area. The archive contains 259 observed monthly solutions from April 2002 to July 2026 relative to the 2004–2009 mean. Source values in centimeters were converted to millimeters, and missing months remain missing. A 12-calendar-month moving mean requires at least nine observations. The 0.25° product grid is finer than the effective GRACE observation scale, so one regional storage indicator is shared across all simulation units (Save, 2020; Save et al., 2016). Soil, surface and other stores were not subtracted; the indicator represents total water storage rather than groundwater alone.

The corresponding aggregation without the CSR land mask is retained as a sensitivity analysis. Regional storage changes provide hydrological context for water-management pressures, with no causal attribution to particular irrigation treatments, water transfers or groundwater regulations (Long et al., 2025; Long et al., 2020).

2.4. Crop model description

Open Crop Model (OCM) couples crop growth and a layered soil-water balance at a daily time step. Wheat development combines temperature, vernalization and photoperiod responses; maize development follows thermal time. Development stage (DVS) is anchored at emergence, flowering and maturity. Germination begins when water above the wilting-point storage is available in the represented layer containing the seed; subsequent thermal development starts from that date. Stage coefficients remain fixed, while the soil-water state can alter establishment timing. Recorded or prescribed harvest dates define the management endpoint separately. Soil-water storage passes continuously through crops and fallows.

Canopy interception of radiation follows Beer–Lambert attenuation. Aboveground production depends on intercepted PAR, crop-specific RUE, temperature, prescribed nutrition and an empirical power response to realized transpiration supply. PAR is 0.45 of incident shortwave radiation (Kiniry et al., 1989). Stage-specific SLA determines new leaf area, while established cohorts retain their birth area-to-mass ratio (de Wit and Berghuijs, n.d.). Stage-dependent thermal senescence and a separate daily stress-loss fraction determine leaf survival (Tan et al., 2020). LAI emerges from these carbon and area balances without a maximum-LAI parameter. Simulated biomass at wheat flowering or maize silking determines grain number; fixed individual grain mass defines sink capacity, and assimilate supply and wheat stem reserves constrain grain filling.

247 The water balance represents interception, ponding, infiltration, evaporation, root uptake, runoff and bottom
248 drainage. Atmospheric demands are partitioned from reference ET using canopy cover and effective transpiration
249 and evaporation coefficients. Soil evaporation follows a storage-based two-stage reduction (Allen et al., 1998).
250 Root uptake depends on rooted-layer water, local depletion stress and a bounded compensation fraction that
251 redirects unmet demand toward better-supplied rooted layers. Vertical transport uses van Genuchten–Mualem
252 hydraulic functions (Mualem, 1976; van Genuchten, 1980), signed gravity–matric gradients and conservative
253 storage bounds. Relative-arithmetic interface mobility and adaptive internal steps permit wetting and redistribution
254 within 24 daily outer steps. The lower boundary is free drainage with no capillary-rise supply. This explicit layered
255 approximation differs from a full implicit Richards-equation solution. Profile depths follow the supplied inputs,
256 including 2 m for Wuqiao and regional simulations.

257 **2.5. Model parameterization, calibration and evaluation**

258 Phenological coefficients, stage-specific specific leaf area and senescence, initial LAI, stem-reserve coefficients and
259 nutrition modifiers follow the crop parameterization described in Section S2. RUE, leaf allocation and grain-
260 number parameters were estimated from 480 wheat and 433 maize growth and harvest observations across four
261 wheat sites and five maize sites. Whole site-years retain their original partitions, with balanced contributions from
262 variables, sites, site-years and crop cases. Station simulations retain their recorded site and cultivar
263 parameterization; Wuqiao and regional simulations use the shared crop parameters (Table S3). Individual grain
264 masses are fixed at 0.045 g for wheat and 0.300 g for maize (APSIM Initiative, 2026).

265 Eight water-use parameters per crop are fitted to the documented Wuqiao treatments in 2016–2018: the
266 transpiration coefficient, soil evaporation coefficient, readily available water fraction, assimilation water-stress
267 exponent, root-density decay, maximum rooting depth, root water-uptake coefficient and root compensation
268 fraction. Each crop contributes 12 seasonal ET, 12 dry-grain and 12 dry-biomass targets, with three annual W3–W0
269 ET contrasts derived from the same treatment seasons. RUE, leaf allocation and grain-number parameters remain
270 fixed. Bounded least squares combines scaled observation residuals and weak parameter priors; objective weights,
271 bounds and convergence settings are specified in Section S2.4.

272 The fitted water parameters are fixed before comparison with the four 2019 treatment means per crop. Previous
273 inspection of these outcomes makes the test retrospective. Station observations have zero weight in this water fit
274 and assess transfer under unconfirmed complete irrigation histories. Regional strategy selection uses historical

weather responses; the crop-calibration years overlap the 2014–2025 regional comparison period, so those outcomes are conditional scenarios rather than time-independent crop validation. Observed and simulated quantities retain matched sampling windows and grain-moisture bases.

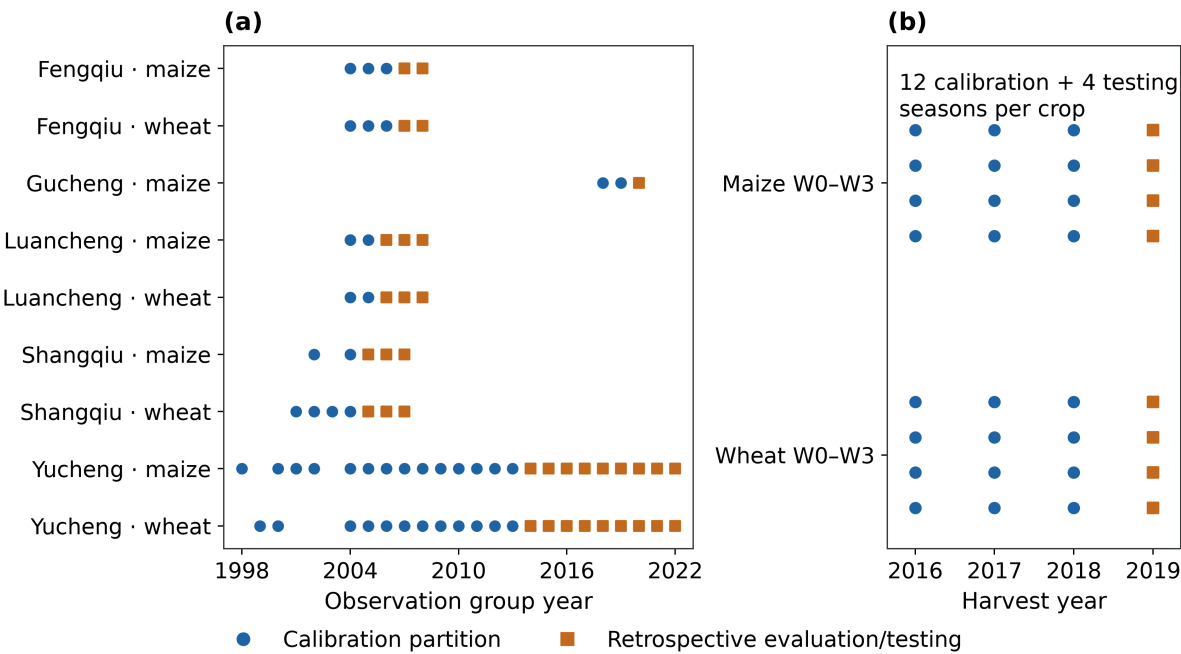


Figure 3. Evidence partitions. (a) Whole station site-years retain their original calibration and retrospective-evaluation partitions; dates denote observation-group years. All five eligible sites are included. (b) Four Wuqiao treatment means per crop-year provide 12 calibration seasons in 2016–2018 and four retrospective testing seasons in 2019. Station comparisons retain earlier growth-fitting history and unconfirmed complete irrigation management.

2.6. Regional upscaling and irrigation scenarios

Thirty-two area-weighted climate–soil representatives were selected from actual source cells using deterministic clustering of latitude, longitude, plant-available water, and 1997–2013 mean precipitation and reference ET. Features were standardized with rotation-area weights. Each cluster uses the actual cell nearest its centroid; the full source-cell mapping and represented areas are retained. Climate years used for policy evaluation do not enter the climate features used for selection. Shared crop parameter sets are transferred regionally without extrapolating site-specific adjustments.

Each representative has five continuous histories at irrigation fractions 0, 0.25, 0.50, 0.75 and 1.00. The conventional wheat schedule is 60 mm at sowing on 12 October, 90 mm on 25 March, 90 mm on 15 April and 60 mm on 10 May, totaling 300 mm. Maize receives 80 mm at sowing on 15 June. These common schedules are declared scenarios and are not observed universal practice across the plain. Every event amount is multiplied by the irrigation fraction, with its timing fixed. Wheat is simulated through 10 June and maize through 5 October;

294 intervening fallows carry the soil state forward. Initial plant-available water is 80% at the start of the 1996 spin-up,
295 with no subsequent seasonal storage reset. All crop and fallow ET, drainage and runoff are included. The final 2025
296 cycle ends on 5 October because the next fallow lies outside the declared endpoint. Nutrition is represented by
297 prescribed fertilized-crop modifiers; nitrogen cycling and spatially varying nutrient limitations are not dynamically
298 simulated.

299 Eight purposively selected non-representative cells provide paired spatial aggregation checks under all five
300 irrigation fractions. Two cells from each latitude quartile have the largest area-times-feature-distance score. Actual-
301 cell weather and profiles use the same continuous model and crop parameters. Their differences from
302 representative-cell predictions measure crop-model aggregation error rather than field-observation accuracy
303 (Supplementary Section S3).

304 **2.7. Irrigation strategies and comparisons**

305 **2.7.1. Spatial allocation under common irrigation budgets**

306 Uniform allocation assigns the same irrigation fraction to every represented hectare. Targeted allocation
307 chooses area shares x_{jk} for representative j and simulated irrigation fraction k by linear programming. The
308 objective maximizes $\sum_j A_j \sum_k x_{jk} \bar{Y}_{jk}$, where $\bar{Y}_{jk}$ is mean annual wheat-plus-maize grain in 1997–2013.
309 Constraints require $\sum_k x_{jk} = 1$, $0 \leq x_{jk} \leq 1$ and $\sum_j A_j \sum_k x_{jk} I_k = (1 - r) \sum_j A_j I_1$, with reductions
310 $r = 0.25, 0.50$ or 0.75 . Within-representative mixtures describe shares of area receiving actual simulated
311 irrigation treatments; crop responses are not interpolated between irrigation fractions. Allocations remain
312 fixed in 2014–2025. Both policies deliver exactly the same total field-irrigation water in every year, so
313 their differences reflect allocation under the same volume constraint.

314 Spatial contrasts at the common 50% irrigation cut use frozen shares of complete continuous irrigation histories.
315 Their area-weighted grain, ET and drainage differences map from 32 representatives to 3,641 source cells. Fixed
316 southern (32–36°N), central (36–38°N) and northern (38–41°N) bands summarize the distribution. Counts of
317 positive annual contrasts describe consistency within the modeled record (Figure 7; Table S9).

318 **2.7.2. Annual quotas based on antecedent water availability**

319 Relative water-availability classes combine two antecedent indicators: regional total-water-storage status and local
320 rainfall before wheat sowing. The storage indicator is the mean of available August and September GRACE
321 observations in the calendar year preceding harvest, with at least one monthly solution required. Precipitation is
322 accumulated from 1 July to 30 September at each simulation unit, covering 92 days before the prescribed 12
323 October wheat sowing. The corresponding precipitation/reference-ET ratio is retained as a diagnostic of climatic
324 water availability. These antecedent indicators do not use rainfall or crop outcomes from the subsequent growing
325 season.

326 Thresholds use 2003–2013 harvest years, the common selection period with antecedent GRACE coverage. The
327 regional storage median is 13.38 mm, and the rotation-area-weighted precipitation median is 431.74 mm. Values
328 below their respective thresholds define lower storage and lower rainfall; values at or above define higher storage
329 and higher rainfall. Crossing the two indicators gives four hydroclimatic classes. The classes describe relative
330 antecedent water availability, rather than absolute water scarcity derived from water demand, allocation rights or
331 access. Spatial differences within a year arise from precipitation; GRACE does not supply independently resolved
332 field-scale storage classes.

333 For each class, five continuous irrigation histories provide paired grain and ET responses relative to conventional
334 irrigation, weighted by rotation area and year. The primary candidate is the lowest fraction retaining at least 95% of
335 class-mean conventional grain in 2003–2013; a 98% target assesses a stricter production criterion. Selected
336 fractions and class thresholds remain frozen in 2014–2025. Higher-storage classes have no evaluation exposure.
337 Required GRACE observations are missing for 2014, 2018 and 2019.

338 Two annual strategies apply the irrigation fractions selected from fixed-fraction responses at grain-retention targets
339 of 95% and 98%. At each simulation unit, the antecedent storage–rainfall class determines the fraction before wheat
340 sowing. This fraction scales all four wheat applications and the subsequent maize application in the same rotation
341 year. Application dates, crop parameters, weather, hydraulic profiles and the free-drainage lower boundary are
342 identical to those in the fixed-fraction experiment. Conventional irrigation applies before 2003 and whenever an
343 eligible storage class is unavailable, including evaluation harvest years 2014, 2018 and 2019.

344 Each strategy is simulated continuously from the 1996 spin-up through 2025. Soil-water states pass between crops
345 and fallows and across annual changes in irrigation fraction. Conventional irrigation supplies the paired continuous
346 reference; replays at two simulation units verify the scenario driver against conventional outputs. Class thresholds
347 and irrigation fractions remain fixed after selection. Outcomes cover all twelve comparison years and, separately,

the nine years with eligible classes. Production retention is the ratio of period-mean regional grain production to its corresponding conventional mean; annual ratios describe variation within the period. The strategies assume access to antecedent observation periods, while release availability of GRACE before sowing remains unverified.

2.7.3. Rainfall-only controls and matched monitoring coverage

Rainfall-only controls use the same 2003–2013 simulation-unit years with eligible GRACE observations, the same rotation-area weights, irrigation candidates and grain-retention targets. Selection groups responses by antecedent rainfall alone and chooses the lowest fraction meeting the relevant target. The rainfall threshold is identical to that in the joint classification. Matched controls receive conventional irrigation whenever the storage–rainfall strategy lacks an eligible class, preserving both the selection cohort and monitoring coverage. Continuous differences between these paired strategies consequently assess the incremental effect of storage-based grouping.

Separate rainfall-available controls apply the selected rainfall-only fractions whenever an eligible rainfall group exists, including years without GRACE coverage. Their complete histories assess the combined effect of rainfall-based management and greater monitoring availability. Irrigation changes in earlier gap years can alter the soil-water state of later crops, so these controls are reported separately from the matched comparison.

2.8. Evaluation metrics, spatial statistics and water accounting

Performance is summarized by RMSE, mean bias error, nRMSE relative to observed mean, Nash–Sutcliffe efficiency (NSE) (Nash and Sutcliffe, 1970) and squared Pearson correlation (R^2). Station scores give equal weight to available sites, site-years within sites, crop cases within site-years and records within cases. Wuqiao treatment means receive equal weights within each crop and partition. Field calibration and retrospective testing remain separate. Grain and harvest biomass in station comparisons are dry matter; Wuqiao grain results retain the published 13% moisture basis, with a single 0.87 conversion between dry and reported grain. Biomass remains dry matter throughout.

Seasonal curves group matching observations and predictions into 15-day wheat and 7-day maize bins at observation-window midpoints in days after sowing. Case means receive equal weight within site-years, site-year means within sites and available sites within bins. Standard errors describe variation among contributing site-year means rather than model uncertainty. Site-specific curves use equal site-year weights. Coverage varies among bins, and gaps remain uninterpolated. Accuracy scores use unbinned records. Yucheng harvest error bars describe quadrat sampling variation within crop-years.

ET sums use identical observed and simulated measurement days within each case and retain the station balancing weights. They exclude unmeasured days and do not represent complete-season observed totals. Regional grain production uses dry mass and rotation-area weights; actual ET includes crops and fallows. Paired policy differences cover 2014–2025. Daily water and carbon residuals and continuity of carried soil-water storage assess numerical conservation. Field irrigation, ET, drainage below 2 m and groundwater storage have distinct accounting boundaries.

Spatial distribution statistics describe targeted-minus-uniform policy contrasts in mean annual combined wheat–maize dry grain and water fluxes over 2014–2025. The 3,641 mapped 0.1° source cells carry outcomes from 32 representative response units. Cell weights are the static 2020 mapped rotation areas after cropland-fraction correction; they represent mapped rotation area rather than observed harvested area. Gains and losses exceed $+10^{-6}$ and fall below -10^{-6} kg ha⁻¹, respectively; contrasts within these bounds are classified as unchanged. Group means use rotation-area weights, regional contributions sum area multiplied by the corresponding contrast, and weighted quantiles use the inverse empirical cumulative area distribution. Annual loss frequencies use the same sign threshold across the 12 years. Connected loss footprints comprise occupied cells sharing a horizontal or vertical grid edge (rook adjacency). The statistics describe the conditional modeled allocation footprint; mapped cells do not provide independent replication or cell-specific validation.

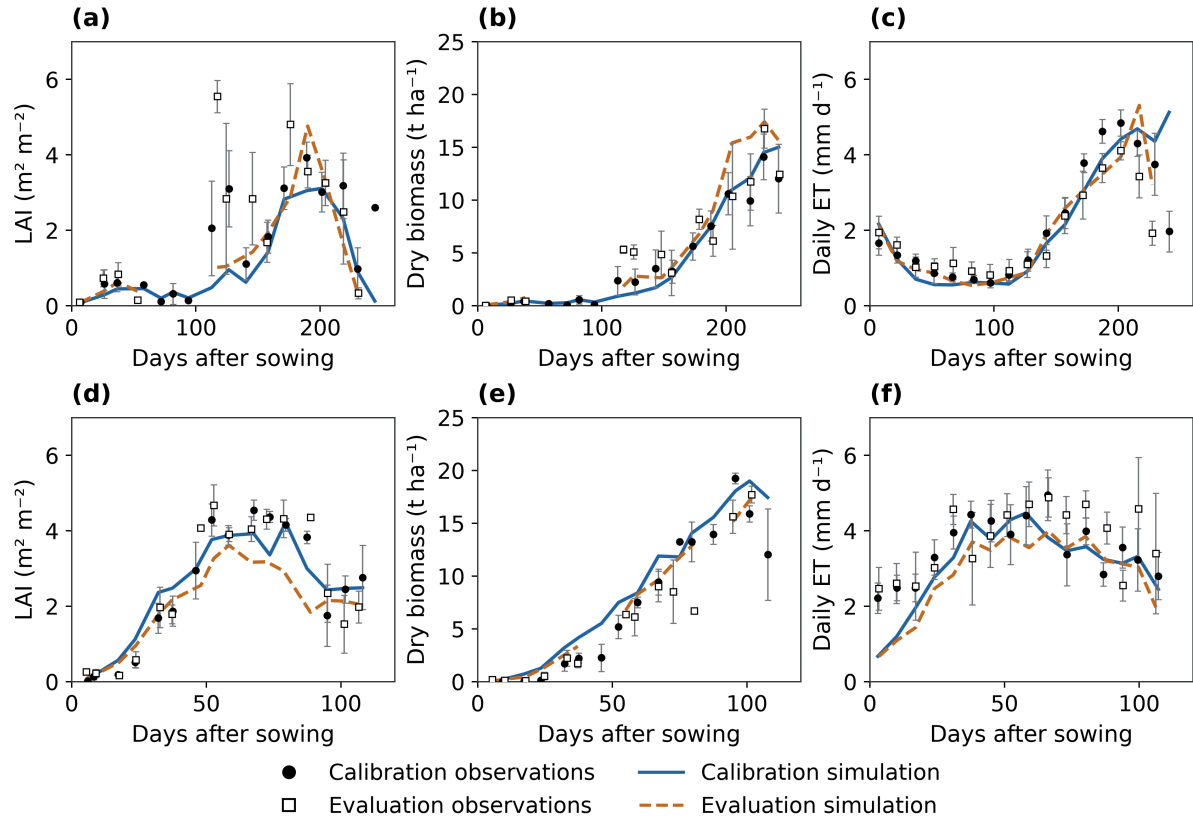
3. Results

3.1. Conditional multisite crop comparisons

Retrospective-evaluation LAI RMSE was 1.843 m² m⁻² for wheat and 0.995 m² m⁻² for maize, with NSE of 0.184 and 0.689, respectively (Figure 4). LAI RMSE in the original calibration partition was 1.727 and 1.138 m² m⁻². In-season biomass RMSE in retrospective evaluation was 4.413 and 2.789 t ha⁻¹, with biases of +1.314 and +1.158 t ha⁻¹ and NSE of 0.349 and 0.796. Site-specific seasonal comparisons retain their available observations and original partitions (Figures S6–S11).

Dry-grain RMSE was 1.256 t ha⁻¹ for wheat and 1.402 t ha⁻¹ for maize in the original calibration partition, compared with 0.834 and 2.286 t ha⁻¹ in retrospective evaluation (Figure S5). Evaluation grain biases were +0.485 and -1.586 t ha⁻¹, and NSE was 0.099 and -1.566. Evaluation harvest-biomass RMSE was 2.736 t ha⁻¹ for wheat and 3.719 t ha⁻¹ for maize; the corresponding observation counts were 54 in nine wheat years and 48 in eight maize years.

404 Daily ET RMSE in retrospective evaluation was 1.809 mm d⁻¹ for wheat and 2.572 mm d⁻¹ for maize, with biases
 405 of +0.141 and -0.734 mm d⁻¹ and NSE of 0.121 and 0.012 (Figure 4). ET summed over matched measurement days
 406 had evaluation RMSE of 51.29 mm for wheat and 74.17 mm for maize, biases of +25.94 and -55.58 mm, and NSE
 407 of 0.826 and 0.118 (Table 2). These sums exclude days without ET measurements.



408 **Figure 4. Multisite seasonal crop comparisons. Wheat and maize occupy the top and bottom rows; columns show LAI, aboveground**
 409 **dry biomass and daily actual ET. Filled and open observations distinguish the original calibration and retrospective-evaluation**
 410 **partitions; solid blue and dashed orange lines show the corresponding simulations. Error bars describe standard errors among site-**
 411 **year means. Balanced means use 15-day wheat and seven-day maize bins, and unobserved bins remain gaps. Accuracy scores use**
 412 **unbinned records. Complete station irrigation management remains unconfirmed.**

413 **Table 2. Conditional station ET sums on matched measurement days.**

Crop	Dataset	Seasons	NSE	RMSE (mm)	MBE (mm)
Maize	Prior calibration partition	11	0.511	45.60	-34.92
Maize	Retrospective evaluation	10	0.118	74.17	-55.58
Wheat	Prior calibration partition	11	-0.300	84.94	-23.53
Wheat	Retrospective evaluation	11	0.826	51.29	25.94

414 Observed and simulated sums use identical measurement days within cases, with balanced site, site-year and case
 415 weights. Unmeasured days are excluded. Original calibration partitions retain an earlier growth-fitting history;
 416 evaluation is retrospective. Complete irrigation-log matching remains unconfirmed.

417 **3.2. Wuqiao calibration and retrospective testing**

418 Seasonal ET calibration RMSE was 40.19 mm for wheat and 37.44 mm for maize across 12 treatment seasons per
 419 crop, with biases of -12.65 and +3.60 mm and nRMSE of 8.97% and 12.69% (Table 3; Figure 5). Across the four
 420 2019 testing means per crop, ET RMSE was 73.80 and 55.64 mm, bias was +48.93 and -35.90 mm, nRMSE was
 421 21.98% and 14.13%, and NSE was 0.503 and -0.346, respectively.

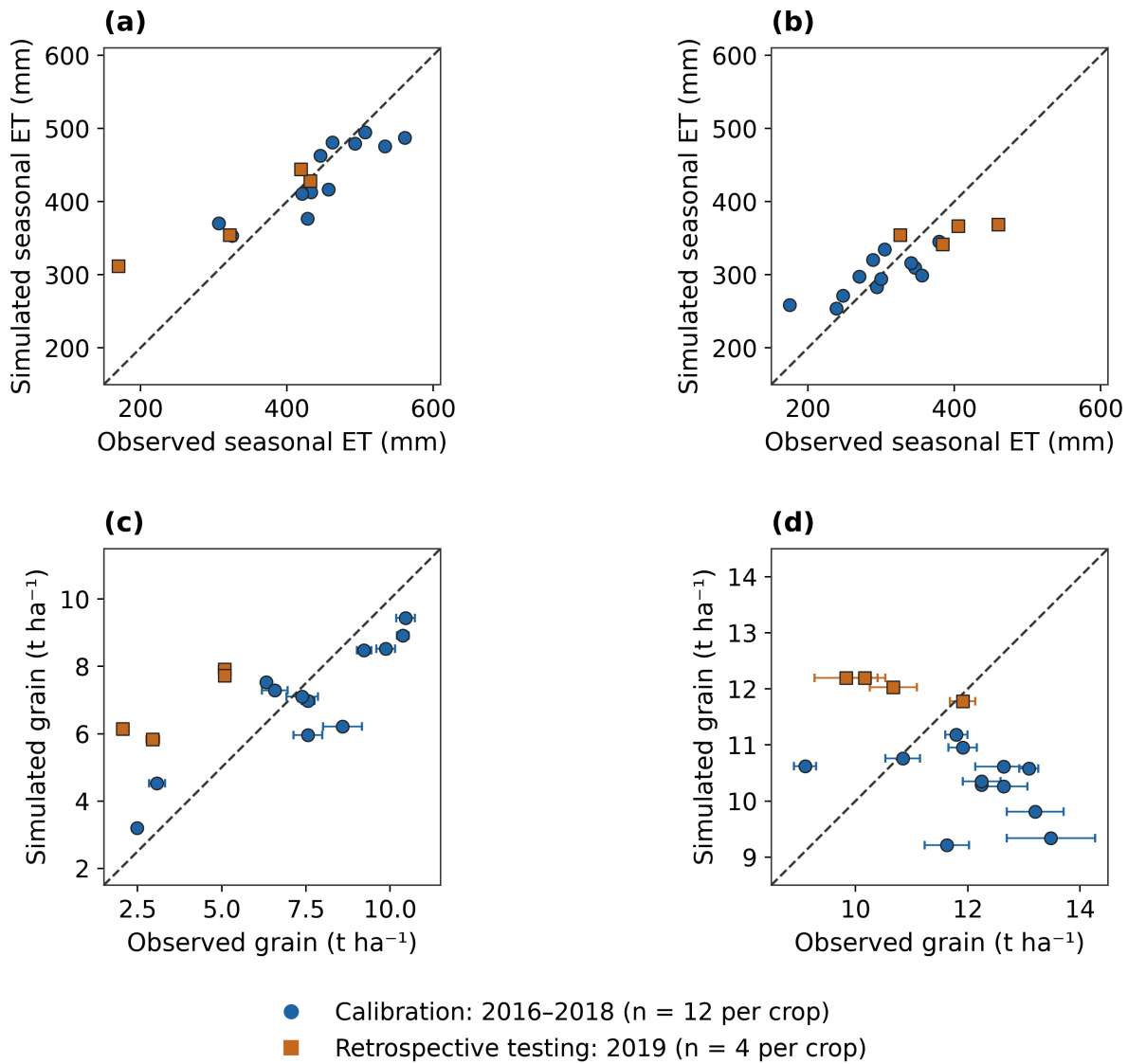
422 At 13% grain moisture, calibration RMSE was 1.249 t ha⁻¹ for wheat and 2.267 t ha⁻¹ for maize; retrospective-
 423 testing RMSE was 3.161 and 1.700 t ha⁻¹. Testing grain biases were +3.108 and +1.403 t ha⁻¹, with nRMSE of
 424 83.27% and 15.97% and NSE of -4.654 and -3.641 (Table 3; Figure 5). Dry aboveground-biomass RMSE was
 425 4.222 and 4.688 t ha⁻¹ in calibration and 5.427 and 2.038 t ha⁻¹ in testing, respectively. Testing biomass nRMSE
 426 was 55.66% for wheat and 10.44% for maize.

427 W3-W0 seasonal ET-contrast RMSE was 45.79 mm for wheat and 66.84 mm for maize across three calibration
 428 years; the absolute 2019 contrast errors were 116.37 and 52.74 mm. For grain contrasts of W1-W3 relative to same-
 429 year W0, calibration RMSE was 0.519 and 1.669 t ha⁻¹ and retrospective-testing RMSE was 1.316 and 1.323 t ha⁻¹,
 430 respectively, at 13% moisture (Table S5; Figure S2). Maize treatment labels denote preceding wheat irrigation;
 431 maize irrigation was identical within each year.

432 **Table 3. Wuqiao water calibration and retrospective testing.**

Crop	Period	Target	n	Unit	RMSE	MBE	nRMSE (%)	NSE	R ²
Maize	2016–2018	Seasonal ET	12	mm	37.444	3.604	12.69	0.533	0.619
Maize	2016–2018	Grain	12	t ha ⁻¹	2.267	-1.738	18.78	-2.940	0.128
Maize	2016–2018	Biomass	12	t ha ⁻¹	4.688	-4.188	21.79	-5.625	0.073
Maize	2019	Seasonal ET	4	mm	55.637	-35.902	14.13	-0.346	0.343
Maize	2019	Grain	4	t ha ⁻¹	1.700	1.403	15.97	-3.641	0.972
Maize	2019	Biomass	4	t ha ⁻¹	2.038	0.726	10.44	-0.313	0.123
Wheat	2016–2018	Seasonal ET	12	mm	40.188	-12.649	8.97	0.689	0.758
Wheat	2016–2018	Grain	12	t ha ⁻¹	1.249	-0.445	16.74	0.746	0.817
Wheat	2016–2018	Biomass	12	t ha ⁻¹	4.222	-2.480	25.48	0.496	0.781
Wheat	2019	Seasonal ET	4	mm	73.805	48.929	21.98	0.503	0.915
Wheat	2019	Grain	4	t ha ⁻¹	3.161	3.108	83.27	-4.654	0.870
Wheat	2019	Biomass	4	t ha ⁻¹	5.427	4.908	55.66	-0.752	0.924

433 Calibration uses 12 treatment seasons per crop in 2016–2018; retrospective testing uses four in 2019. Counts are
434 treatment means rather than independent replicate seasons. ET is a 0–2 m water-balance estimate under negligible
435 runoff and drainage assumptions. Grain is at 13% moisture and aboveground biomass is dry matter. Annual grain
436 means and SE are digitized from source Supplementary Figure S2. Initial total storage is observed; layer
437 distribution and transferred hydraulics remain conditional.



438 **Figure 5.** Wuqiao calibration and retrospective testing. (a,b) Seasonal ET; (c,d) annual treatment-mean grain at 13% moisture.
 439 **Wheat occupies the left column and maize the right. Circles show 2016–2018 treatments used to fit water-use parameters; squares**
 440 **show 2019, withheld from the fit and selection. Dashed lines denote 1:1 agreement. Crop-specific grain axes focus on their observed**
 441 **ranges. Horizontal grain error bars are the published SE from three field replicates. Observed ET follows the published 0–2 m water**
 442 **balance; simulated dry grain is converted once by division by 0.87. Maize labels retain preceding wheat management with identical**
 443 **maize irrigation within each year.**

444 3.3. Grain production and water balance under irrigation allocation

445 At a 50% regional irrigation reduction, targeted allocation gave a simulated annual rotation dry-grain yield of
 446 15.761 t ha⁻¹ yr⁻¹ for wheat and maize combined, compared with 15.029 under uniform cuts. Both yields remained
 447 below the conventional yield of 16.756 t ha⁻¹ yr⁻¹. Targeting therefore retained 0.732 t ha⁻¹ yr⁻¹ more grain than

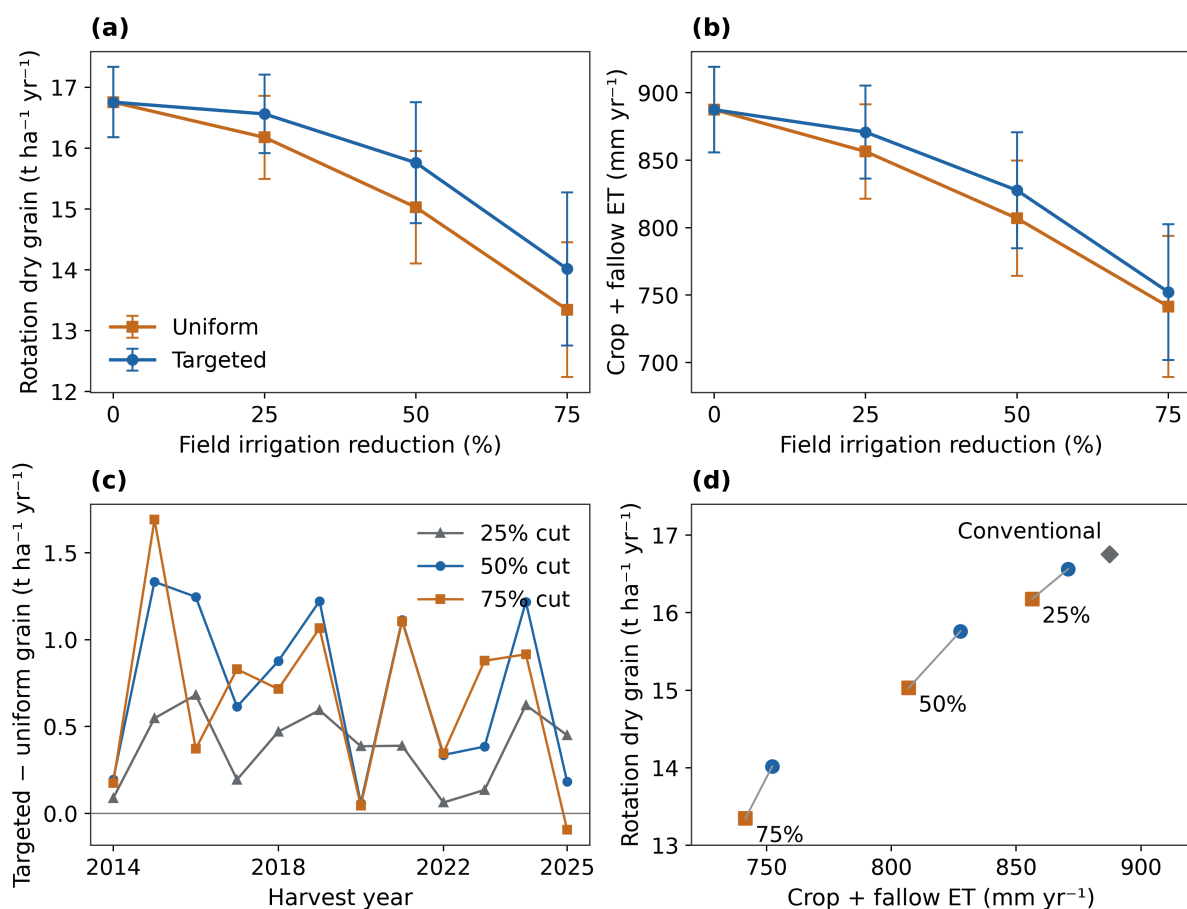
uniform allocation (4.87% of uniform production), equivalent to 7.219 million t yr⁻¹ across the fixed 9.867 million rotation hectares (Table 4; Figure 6). Targeted production remained 9.813 million t yr⁻¹ below conventional production.

Both allocations supplied 190 mm yr⁻¹ of field irrigation. Targeted and uniform crop-plus-fallow ET averaged 827.70 and 806.94 mm yr⁻¹, respectively. The grain advantage was accompanied by 20.76 mm yr⁻¹ (2.048 km³ yr⁻¹) greater ET. Profile drainage was 79.04 versus 96.34 mm yr⁻¹, and runoff was 14.43 versus 18.99 mm yr⁻¹ (Table 4). The targeted-minus-uniform grain contrast at the 50% cut was positive in all twelve comparison years (Figure 6c). The positive regional contrast was accompanied by spatially contrasting yield responses (Section 3.4).

Table 4. Simulated annual rotation production and water balance under equal irrigation reductions, 2014–2025.

Strategy	Dry grain (t ha ⁻¹ yr ⁻¹)	Irrigation (mm yr ⁻¹)	ET (mm yr ⁻¹)	Drainage (mm yr ⁻¹)	Runoff (mm yr ⁻¹)
Conventional	16.756	380.00	887.43	185.26	39.32
Uniform, 25% cut	16.177	285.00	856.41	135.94	25.71
Targeted, 25% cut	16.562	285.00	870.81	125.90	19.86
Uniform, 50% cut	15.029	190.00	806.94	96.34	18.99
Targeted, 50% cut	15.761	190.00	827.70	79.04	14.43
Uniform, 75% cut	13.347	95.00	741.57	68.72	15.83
Targeted, 75% cut	14.018	95.00	752.30	60.01	13.68

All per-hectare means use the same 9.866602 million mapped rotation hectares. Dry grain combines wheat and maize harvests; ET includes crops and fallows. Regional grain in million tonnes equals rotation yield multiplied by mapped area in million hectares. Regional water in km³ equals depth in mm multiplied by hectares and 10, divided by 10⁹. The conventional schedule supplies 300 mm to wheat and 80 mm to maize.



461 **Figure 6. Irrigation allocation and grain–water trade-offs during 2014–2025. (a,b) Rotation-area-weighted annual combined wheat–**
 462 **maize dry grain and crop-plus-fallow ET; error bars are between-year SD. (c) Annual targeted-minus-uniform grain at identical**
 463 **irrigation budgets. (d) Period means; connectors join policies with the same budget. Values per hectare use the same 9.867 million**
 464 **mapped rotation hectares, rather than separate crop areas. Targeting retains more grain than uniform cuts while remaining below**
 465 **conventional production.**

466 3.4. Spatial distribution and persistence of yield contrasts

467 Relative to uniform allocation, the central and northern latitude bands contributed grain gains of 7.475 and 1.886
 468 million $t\ yr^{-1}$, while the southern band contributed a loss of 2.142 million $t\ yr^{-1}$. The southern band occupied 60.7%
 469 of mapped rotation area. Mean rotation-yield contrasts were +2.415, +2.420 and $-0.357\ t\ ha^{-1}\ yr^{-1}$ in the central,
 470 northern and southern bands, respectively (Figure 7; Table S9).

471 At the 50% regional irrigation reduction, targeted allocation increased mean annual combined wheat–maize dry
 472 grain relative to uniform reduction on 43.92% of the mapped rotation area (4.333 Mha), decreased it on 37.60%
 473 (3.710 Mha), and left it unchanged on 18.48% (1.823 Mha). Mean contrasts within the gain and loss areas were

474 +2.439 and -0.903 t ha⁻¹ yr⁻¹, respectively. Their regional contributions were +10.568 and -3.349 Mt yr⁻¹, giving a
475 net regional grain difference of 7.219 Mt yr⁻¹ (0.732 t ha⁻¹ yr⁻¹ across the full mapped area). Local losses offset
476 31.69% of the gross positive contribution (Figure 7d).

477 The area-weighted median grain contrast was 0.000 t ha⁻¹ yr⁻¹, and the 10th and 90th percentiles were -1.182 and
478 +3.553 t ha⁻¹ yr⁻¹. Relative to uniform reduction, mean annual field irrigation increased by 119.0 mm yr⁻¹ in the
479 gain area and decreased by 139.0 mm yr⁻¹ in the loss area. The corresponding crop-plus-fallow ET contrasts were
480 +94.4 and -55.1 mm yr⁻¹. Irrigation changes balanced at regional scale, whereas the area-weighted mean ET
481 contrast was +20.8 mm yr⁻¹.

482 Negative yield contrasts persisted over 2014–2025: 21.34% of the mapped rotation area had lower yields than under
483 uniform allocation in all 12 years, and 37.60% experienced a loss in at least one year. Every response unit with a
484 negative period-mean contrast recorded losses in at least 9 years; all response units with a positive period-mean
485 contrast recorded gains in all 12 years. Across years, the mean annual area exposed to a loss was 35.37%. Mapped
486 cells with negative period-mean contrasts formed 8 connected components under rook adjacency. The largest
487 contained 3.700 Mha of mapped rotation area (99.73% of all loss area), with occupied cell centers extending from
488 32.0 to 36.3°N and 112.8 to 121.5°E.

489 Negative period-mean yield contrasts occupied 32.80%, 37.60%, and 60.08% of mapped rotation area at reductions
490 of 25%, 50%, and 75%, respectively. At the 75% reduction, the area-weighted median contrast became negative
491 (-0.788 t ha⁻¹ yr⁻¹), although gains on 37.04% of the area produced a positive regional balance. Gross positive and
492 negative contributions at that reduction were +14.100 and -7.487 Mt yr⁻¹, leaving a net gain of 6.614 Mt yr⁻¹, below
493 the net gain at the 50% reduction.

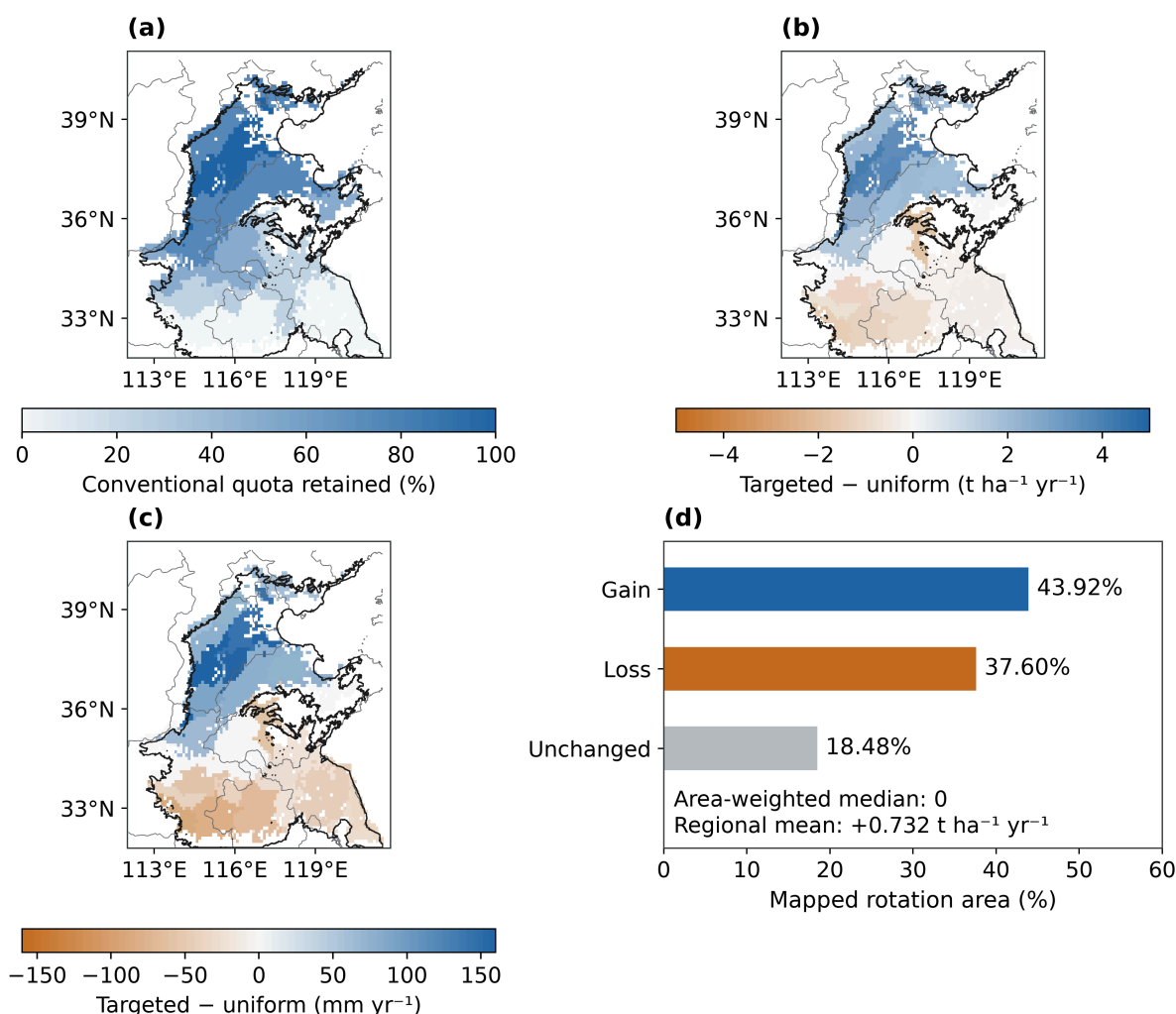


Figure 7. Spatial redistribution at a common 50% regional irrigation reduction, 2014–2025. (a) Targeted fraction of the conventional quota; uniform allocation retains 50% everywhere. (b,c) Targeted-minus-uniform combined rotation dry grain and crop-plus-fallow ET. (d) Mapped rotation-area shares with positive, negative or unchanged mean grain contrasts. The 3,641 cells inherit 32 representative responses. Thin grey provincial lines and black study-domain outlines follow Figure 2; unmapped areas remain blank.

3.5. Antecedent water availability and annual irrigation strategies

Four antecedent storage–rainfall classes occurred during 2003–2013. Nine comparison years had eligible GRACE observations, all in the lower-storage category; three years without eligible GRACE observations received conventional irrigation quotas. At the 95% grain target, both storage strata selected an irrigation fraction of 0.75 under lower rainfall and 0.50 under higher rainfall. The 98% target selected 1.00 and 0.75, respectively (Table 5). Continuous strategies selected at the 95% and 98% targets retained 96.27% and 99.63% of conventional grain production over the nine years with eligible GRACE observations. Field-irrigation reductions were 130.72 and 35.72 mm per rotation; corresponding ET reductions were 36.84 and 6.17 mm (Figure 8; Table S7). The 95% strategy retained at least 95% in 8 of nine years, with a minimum of 92.12%.

507 Storage–rainfall and matched rainfall-only strategies selected identical irrigation fractions and produced identical
 508 grain, ET, drainage and irrigation histories at both targets. Rainfall-only strategies that applied decisions during
 509 GRACE gaps followed different irrigation histories and carried different soil-water states into subsequent crops
 510 (Table S8).

511 **Table 5. Selected irrigation candidates by relative water-availability class.**

Class	Irrigation fraction	Wheat irrigation (mm)	Maize irrigation (mm)	Selection grain retention (%)
Higher storage dry	0.750	225.000	60.000	95.537
Higher storage wet	0.500	150.000	40.000	97.093
Lower storage dry	0.750	225.000	60.000	96.617
Lower storage wet	0.500	150.000	40.000	97.431

512 Fractions meet the 95% class-mean grain target during selection years 2003–2013. Current shared crop parameters
 513 apply throughout. Both storage strata select identical fractions within each rainfall group; higher-storage classes
 514 have no later-period exposure.

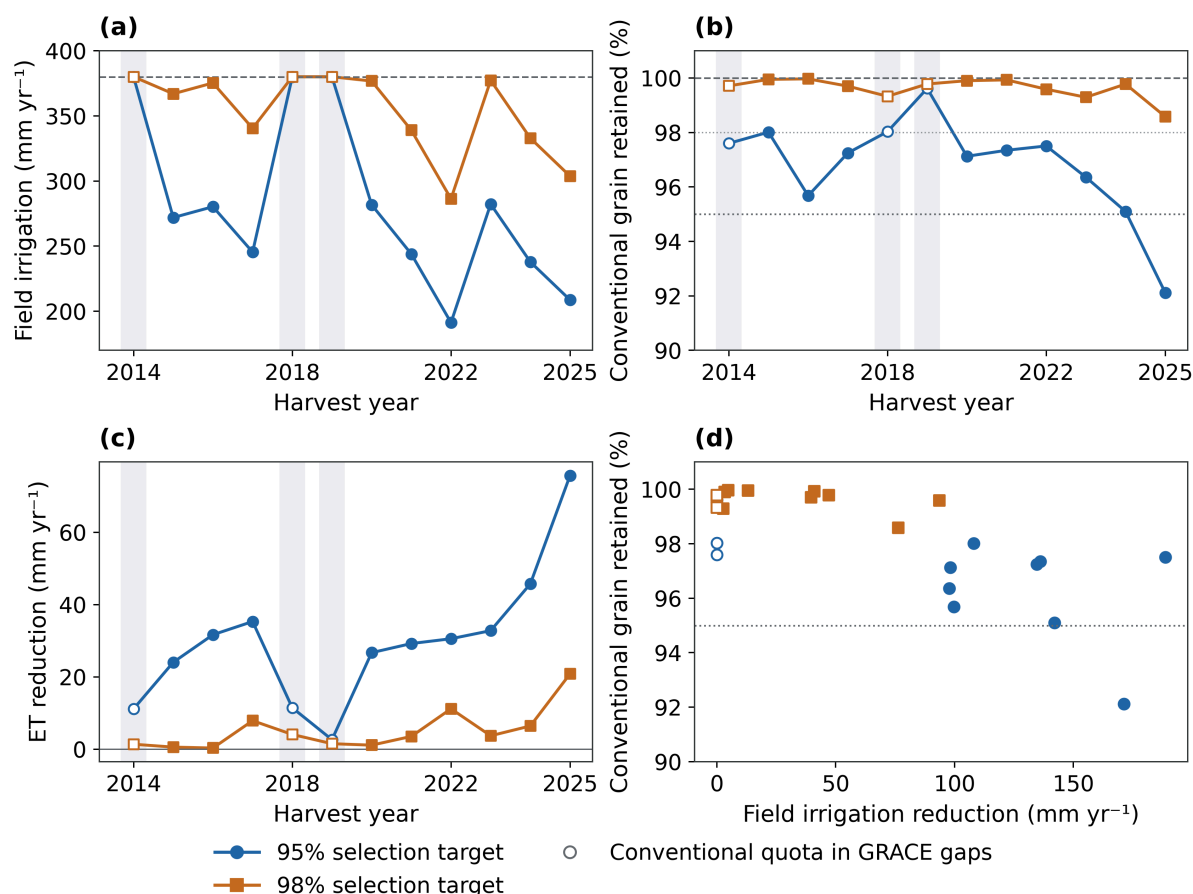


Figure 8. Continuous annual irrigation strategies during 2014–2025. (a) Field irrigation; (b) grain retained relative to conventional irrigation; (c) crop-plus-fallow ET reduction; (d) annual production–irrigation combinations. Blue circles and orange squares identify 95% and 98% selection targets. Shading and open markers identify missing-GRACE years with conventional quotas. Grain and ET can still differ in those years because preceding strategy decisions alter carried soil-water states. Matched rainfall-only histories coincide with the storage–rainfall histories.

4. Discussion

The simulations reveal three linked dimensions of irrigation management: production retained under a delivery constraint, consumptive water use and the distribution of production changes. Spatial targeting improves regional grain production by assigning a fixed budget to more responsive crop–soil–climate systems. Annual rainfall–storage strategies instead change the delivered quota in response to antecedent conditions. Both decisions operate through crop stress and the soil water carried between wheat, maize and fallow periods. Their management value consequently depends on the complete rotation response, the accounting boundary used to define a water saving and the reliability of the crop responses underlying selection.

4.1. Marginal irrigation returns and spatial allocation

Spatial targeting exploits differences in marginal grain response to irrigation. Diminishing production gains across field schedules (Yang et al., 2022) and spatial variation in irrigation requirements (Wu et al., 2019) provide a basis for redistributing a limited supply, consistent with regional allocation studies (Jiang et al., 2016). The equal-budget comparison attributes the simulated production difference to delivery allocation. Complete continuous histories retain the effects of earlier irrigation and rainfall on subsequent crop responses.

At the 50% delivery reduction, targeting retained $0.732 \text{ t ha}^{-1} \text{ yr}^{-1}$ more combined wheat–maize dry grain than uniform cuts. Across 9.8666 million rotation hectares, the corresponding reduction in grain loss was 7.219 million t yr^{-1} , or 4.87% of the uniform total. Targeted production remained below conventional production. The positive contrast in all twelve years indicates consistent regional gains within the simulated record. Its magnitude remains dependent on the crop-response functions and declared management schedules (Table 4; Figure 6).

The spatial distribution sharpens this distinction between regional and local outcomes. Relative to uniform allocation, mean losses occupy 37.60% of mapped rotation area, and losses persist in every comparison year on 21.34%. Gross gains of $10.568 \text{ Mt yr}^{-1}$ are partly offset by losses of 3.349 Mt yr^{-1} ; the area-weighted median is zero despite a positive regional mean. The regional grain objective balances gains and losses among locations. Local production floors constitute an additional allocation constraint.

Regional gain has a pronounced distributional cost. Central and northern band-mean rotation yields increased by 2.415 and $2.420 \text{ t ha}^{-1} \text{ yr}^{-1}$, while the southern mean decreased by $0.357 \text{ t ha}^{-1} \text{ yr}^{-1}$. The southern band covers 60.7% of the rotation area, so its losses substantially offset gains elsewhere (Figure 7; Table S9). The production objective maximizes the area-weighted regional total and supplies no minimum yield for an individual location. Local production floors and constraints on the distribution of irrigation reductions are therefore separate requirements for allocation that protects affected farms.

4.2. Rotation carry-over and the timing of water supply

The spatial production response integrates water effects across both crops. Wheat irrigation can support canopy growth and grain formation, replenish water available to maize or increase drainage. Reduced wheat delivery can increase subsequent dependence on summer rainfall, producing different annual grain responses under different rainfall sequences. The sixteen-year irrigation experiment shows diminishing production gains, and the Wuqiao experiment links precipitation conditions with wheat scheduling and subsequent maize production (Yang et al., 2024; Yang et al., 2022). These coupled responses make the complete rotation the relevant unit for irrigation allocation.

Continuous soil-water histories preserve the delayed benefits and costs of allocation and annual quota changes (Figure S3). Production can be maintained by drawing down profile storage, with later rainfall or irrigation determining its replenishment. A finite-period delivery reduction can therefore include a contribution from stored soil water. Crop-plus-fallow accounting also captures evaporation and drainage between harvest and sowing. Their inclusion connects apparent production retention with the water remaining for the next crop and the complete rotation water balance.

The field evidence constrains only part of this carry-over mechanism. Wuqiao simulations use reported total profile storage with an assumed layer distribution. Wheat crop-start storage follows the pre-sowing simulation, whereas maize is independently initialized from treatment-specific observations. Maize irrigation is identical among treatments within each year, and its treatment labels identify preceding wheat schedules (Yang et al., 2024). Maize comparisons thus evaluate crop responses to the initial state associated with wheat management (Figure 5; Figure S2). Regional carry-over additionally depends on the model's continuous propagation and vertical redistribution of water between crops.

4.3. Production, consumptive use and groundwater accounting

The production benefit of targeting carries a consumptive-use trade-off. At the common 50% delivery reduction, targeting increased ET by $2.048 \text{ km}^3 \text{ yr}^{-1}$ and reduced profile drainage by $1.707 \text{ km}^3 \text{ yr}^{-1}$ relative to uniform allocation. With rainfall and total irrigation held constant, differences in ET are balanced by drainage, runoff and soil-storage change (Table 4). A shared delivery constraint therefore permits different consumptive outcomes. Irrigation productivity and consumptive-water savings require distinct objectives (Grafton et al., 2018; Perry et al., 2009).

The soil-profile balance also has a different boundary from the aquifer balance. Drainage below 2 m enters a deeper vadose zone, with storage and transit outside the modeled profile (Wu et al., 2023). Reduced field delivery or profile drainage therefore provides insufficient information to quantify changes in pumping, aquifer recharge or groundwater storage. Production and ET comparisons can identify agronomic trade-offs, while groundwater outcomes require the corresponding subsurface and abstraction accounting.

GRACE offers regional storage context at a broader scale. Its anomalies integrate groundwater with other terrestrial stores, and its effective observation scale exceeds the displayed grid resolution (Save et al., 2016; Scanlon et al., 2016). Recent aquifer recovery reflects water transfers, abstraction management and hydroclimatic conditions

586 (Long et al., 2025; Xu et al., 2026). The crop-model scenarios quantify changes in field delivery, consumption and
587 profile outflow without attributing those observed storage changes to individual irrigation strategies.

588 **4.4. When storage information changes an irrigation decision**

589 A storage indicator adds decision value when its inclusion changes the selected action or improves the grain–water
590 outcome under common constraints. GRACE represents regional storage conditions, while antecedent rainfall
591 describes more local water input (Save et al., 2016; Scanlon et al., 2016). Their different physical support motivates
592 joint classification, but decision value also depends on how the resulting classes distinguish among the available
593 irrigation fractions.

594 Both storage strata selected the same fraction for a given rainfall group at the 95% and 98% grain targets. Storage–
595 rainfall and matched rainfall-only strategies consequently produced identical complete histories of irrigation, crop
596 states, grain, ET and drainage. The storage split supplied additional environmental description without changing an
597 irrigation action in this five-level quota design. All nine later years with eligible observations fell in the lower-
598 storage category, so the higher-storage choices also lacked later-period exposure (Table 5; Figure 8). These features
599 delimit the configuration and conditions represented by the equal outcomes.

600 Monitoring coverage must remain distinct from the grouping criterion. Rainfall-available controls can change
601 irrigation during GRACE gaps, and those earlier decisions alter the soil-water state inherited in later years. Their
602 outcomes therefore combine rainfall-based selection with additional years of management. Matched coverage
603 preserves the appropriate comparison for the incremental effect of storage information. Finer quota choices,
604 alternative thresholds and verified release availability would define different decision experiments with their own
605 selection and evaluation requirements.

606 **4.5. Crop-response constraints and management reliability**

607 Management selection depends on the modeled marginal response to water, in addition to absolute grain and canopy
608 agreement. Intercepted-PAR radiation-use efficiency (Kiniry et al., 1989), cohort-specific leaf area and thermal
609 senescence connect canopy development with carbon supply; grain number and fixed individual grain mass
610 constrain the sink. Root uptake and the assimilation stress exponent then determine how water shortage changes
611 production. Variation among wheat-model transpiration formulations demonstrates the sensitivity of water-use
612 predictions to these processes (Cammarano et al., 2016). Water coefficients near their prescribed bounds remain
613 effective empirical estimates whose physiological interpretation and uniqueness are limited.

614 The estimation and evaluation data provide different evidence for these responses. Earlier growth estimates use 480
615 wheat and 433 maize observations across four wheat sites and five maize sites, with whole site-years and balanced
616 site contributions. Incomplete station irrigation histories limit transfer comparisons, and station observations have
617 no weight in the fitted water-response objective. Wuqiao 2016–2018 supplies the documented treatment seasons for
618 that fit. Wuqiao 2019 supplies a single test year with reported total storage, assumed layer distributions and
619 transferred soil hydraulics; prior inspection of its outcomes makes the test retrospective (Figures 3–5). Regional
620 comparison years also overlap crop-fitting years, so regional production differences characterize model scenarios
621 rather than time-forward crop validation.

622 The absolute regional production reference also depends on the prescribed management and crop parameters.
623 Conventional simulations produce 8.065 t ha^{-1} of wheat dry grain and 8.691 t ha^{-1} of maize dry grain per year.
624 These values represent fixed irrigation dates, crop calendars and nutrition modifiers rather than an independently
625 validated regional harvest inventory. The $0.732 \text{ t ha}^{-1} \text{ yr}^{-1}$ allocation advantage depends on the same simulated
626 marginal responses; its regional scaling does not remove field-response uncertainty.

627 Substantial field-test errors constrain confidence in irrigation-response transfer. Wheat grain nRMSE was 83.27%,
628 with a bias of $+3.108 \text{ t ha}^{-1}$ at the published 13% moisture basis; the absolute W3–W0 ET-contrast error was 116.37
629 mm. Maize ET NSE was negative, with an ET-contrast error of 52.74 mm (Table 3; Table S5). These discrepancies
630 affect the response functions used to select regional allocations. Numerical water-balance closure demonstrates
631 conservation but supplies no measure of response accuracy. The eight purposive actual-cell checks address a
632 separate uncertainty, spatial aggregation, and provide no regional field-validation sample. Depth-resolved soil water
633 and complete dated management would better constrain uptake and redistribution; canopy and soil-moisture
634 assimilation also requires treatment of retrieval errors and observation operators (Ines et al., 2013).

635 Production-retention targets introduce a further management distinction between period means and annual
636 reliability. Mean retention of 96.27% and 99.63% coincided with annual minima of 92.12% and 98.58%; the 95%
637 strategy met that threshold in eight of nine class-available years (Figure 8). A mean target consequently supplies no
638 shortfall probability or guarantee for an individual farm. Local production floors, acceptable shortfall frequency and
639 a reserve entering the next crop can complement regional delivery and consumption constraints. Forecast-based
640 irrigation studies illustrate explicit treatment of water-stress risk (Roy et al., 2021). Field scheduling additionally
641 requires current soil-water information, weather forecasts and application timing.

642 **5. Conclusions**

643 At an identical 50% irrigation reduction, targeted allocation retained 0.732 t ha⁻¹ yr⁻¹ more simulated combined
644 wheat–maize dry grain than uniform cuts during 2014–2025. This reduced regional grain loss by 7.219 million t
645 yr⁻¹ (4.87% of uniform production), while increasing crop-plus-fallow ET by 20.76 mm yr⁻¹ (2.048 km³ yr⁻¹).
646 Relative to uniform allocation, mean grain losses occurred on 37.60% of mapped rotation area and persisted in all
647 twelve years on 21.34%. Regional production retention therefore requires separate consumption and local
648 production constraints.

649 Annual strategies selected at 95% and 98% grain-retention targets retained 96.27% and 99.63% of conventional
650 production over nine years with eligible GRACE observations. The 95% strategy reduced irrigation by 130.72 mm
651 and ET by 36.84 mm per rotation. Matched rainfall-only controls selected identical quotas and produced identical
652 outcomes, so storage classification provided no additional decision benefit within the tested quota design and
653 monitoring coverage.

654 Continuous rotation accounting preserves the influence of preceding management on subsequent production and
655 water use. The field-test errors and incomplete station management delimit confidence in transferring the estimated
656 crop responses. Translation of these regional scenarios into irrigation management depends on observed response
657 accuracy, explicit production-risk constraints and field scheduling informed by soil water and forecasts.
658 Groundwater effects additionally require pumping, recharge and deeper subsurface storage beyond the crop-model
659 boundary.

660 **Data and code availability**

661 Model code, frozen calibration and regional inputs, observation subsets and quality exclusions, weather, calibrated
662 parameters, predictions, evaluation metrics and figure-data tables are available at [https://github.com/SmartAG-](https://github.com/SmartAG-Team/agri_water_management)
663 [Team/agri_water_management](https://github.com/SmartAG-Team/agri_water_management). The repository includes a standalone Open Crop Model package, pinned
664 dependencies, reproduction commands and SHA-256 provenance. Reproduction uses the included inputs without
665 another model checkout or access to the historical workspace. Original datasets retain their provider licences and
666 attribution requirements, documented in the repository source catalogue.

667 **Declaration of generative AI and AI-assisted technologies**

668 AI assisted with grammar checking.

669 **Supplementary material**

670 **Supplementary methods and results**

671 **S1. Observation sources and eligibility**

672 The station archive preserves 132 crop–plot–season trajectories, whole site-year partitions and source metadata. The
673 7,397 eligible records occur in 127 cases and retain all five observation sites for their available variables.

674 Calibration and retrospective-evaluation trajectory counts remain 71 and 61. Variable-specific availability and
675 missing combinations are retained in the accompanying case and site tables; missing measurements do not become
676 zero observations.

677 Wheat LAI above $7 \text{ m}^2 \text{ m}^{-2}$ remains excluded under the study-specific rule. Duplicated six-replicate maize harvest
678 vectors in Yucheng 2012 and 2017 account for 24 excluded grain and biomass records, divided equally between the
679 original calibration and evaluation partitions. These source-based exclusions retain raw values and do not depend
680 on prediction residuals.

681 Earlier multisite growth estimates retain 480 wheat and 433 maize calibration observations. Current station growth
682 and ET comparisons are conditional on archived management whose full irrigation-log completeness is unresolved.
683 They have zero weight in the water-parameter fit. Wuqiao 2016–2018 supplies 12 documented treatment seasons
684 per crop for that fit (Yang et al., 2024); 2019 supplies four retrospective testing seasons per crop. Seasonal ET, grain
685 and biomass targets refer to the same treatment-season population, while ET contrasts are derived from those
686 seasonal targets.

687 Twenty-six Fengqiu cases retain antecedent moisture initialization from the preceding calendar month. Complete
688 fifteen-layer 0–1.5 m volumetric profiles map conservatively to the represented model layers, preserving depth-
689 integrated storage. The identical eligibility rule applies to 11 original calibration and 15 evaluation cases. Monthly
690 timing, aggregation and subplot support remain unresolved relative to the exact sowing-day state. Original cases
691 remain available when a measured antecedent profile is absent.

692 **Table S1. Crop seasons and calibration/evaluation case counts by site.**

Site	Crop	Years	Calibration cases	Evaluation cases
Fengqiu	Maize	2004, 2005, 2006, 2007, 2008	11	9
Fengqiu	Wheat	2004, 2005, 2006, 2007, 2008	11	10
Gucheng	Maize	2018, 2019, 2020	8	4
Luancheng	Maize	2004, 2005, 2006, 2007, 2008	2	3
Luancheng	Wheat	2004, 2005, 2006, 2007, 2008	2	3
Shangqiu	Maize	2002, 2004, 2005, 2006, 2007	2	6
Shangqiu	Wheat	2001, 2002, 2003, 2004, 2005, 2006, 2007	9	8
Yucheng	Maize	1998, 2000, 2001, 2002, 2004, 2005, 2006, 2007, 2008, 2009, 2010, 2011, 2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022	14	9
Yucheng	Wheat	1999, 2000, 2004, 2005, 2006, 2007, 2008, 2009, 2010, 2011, 2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022	12	9

693 Years denote archived crop-season groups, which retain all their plot cases. Coverage is intermittent; years absent
694 from the table are not assumed to have observations.

695 **Table S2. Source datasets for evaluated field observations.**

Site	Dataset	DOI	Source
Fengqiu	2004-2008年河南封丘站	10.12199/nesdc.ecodb.mon.	National Ecosystem
	作物生育及生长动态监测	2020.dp2011.fqa.002	Science Data Center /
	数据		Science Data Bank
Fengqiu	2005-2008年河南封丘站	10.12199/nesdc.ecodb.mon.	National Ecosystem
	农田土壤水分物理性质监	2020.dp2011.fqa.006	Science Data Center /
	测数据		Science Data Bank
Gucheng	2018-2021年华北北部不	10.57760/sciencedb.o00119	National Ecosystem
	同播期对玉米生长发育的	.00017	Science Data Center /
	影响数据集		Science Data Bank
Luancheng	2000-2008年河北栾城站	10.12199/nesdc.ecodb.mon.	National Ecosystem
	作物生育及生长动态监测	2020.dp2011.lca.002	Science Data Center /
	数据		Science Data Bank
Shangqiu	2000-2007年河南商丘站	10.12199/nesdc.ecodb.mon.	National Ecosystem
	作物生育及生长动态监测	2020.dp2011.sqa.003	Science Data Center /
	数据		Science Data Bank
Yucheng	1998-2006年山东禹城站	10.12199/nesdc.ecodb.mon.	National Ecosystem
	作物生育及生长动态监测	2020.dp2011.yca.003	Science Data Center /
	数据		Science Data Bank
Yucheng	禹城站2005-2022年生物	10.12199/nesdc.ecodb.2021	National Ecosystem
	要素观测数据集	YFF0703900.yca.2025.2	Science Data Center /
			Science Data Bank
Yucheng	禹城站2005-2022年水环	10.12199/nesdc.ecodb.2021	National Ecosystem
	境要素观测数据集	YFF0703900.yca.2025.3	Science Data Center /
			Science Data Bank

696 Dataset DOIs identify official source records; original metadata and observation-row provenance are retained in the
697 reproducibility archive.

698 **S2. Crop processes, parameter definitions and water accounting**

699 **S2.1. Model structure and phenological development**

Open Crop Model simulates phenology, green leaf area, live and dead aboveground biomass, grain dry matter, rooting depth and layered soil-water storage. Inputs comprise weather, soil profiles, crop parameters, dated sowing and harvest, and irrigation events. Soil water carries continuously between crop and fallow segments in regional histories; Wuqiao crop starts retain their separately specified storage conditioning.

Wheat stage development combines temperature, vernalization and photoperiod modifiers with inherited stage requirements. Maize uses a subdaily temperature response reconstructed from daily extrema, with base, optimum and maximum temperatures of 6, 30 and 44 °C. The shared medium-maturity maize card requires approximately 1,909 °C d to reach R6. DVS anchors are wheat BBCH 09, 61 and 89 and maize VE, R1 and R6 (Meier, 2018).

Wheat dough stage and full ripeness retain their distinct definitions.

Seed-layer water above wilting-point storage permits germination. The represented layer is selected by sowing depth, with a 50-mm prior when no depth is supplied. Thermal development and wheat vernalization begin from the first eligible germination day rather than accumulating during dry-seed days. This layer-average availability is an effective establishment condition, not a measured seed-scale water potential. Fixed stage coefficients therefore do not imply fixed calendar-stage predictions across water states. New biomass production ends at wheat full ripeness or maize physiological maturity; green maize area can persist under the terminal senescence rate until harvest. Harvest is a separate management endpoint and does not substitute for an observed maturity date.

S2.2. Canopy development, biomass accumulation and grain yield

Green cover is $f_c = 1 - \exp(-k_L \text{LAI})$. Daily aboveground dry-matter production is $\Delta B = 10R_s p_{\text{PAR}} f_c \varepsilon f_T f_W^y f_N$, where R_s is incident shortwave radiation, $p_{\text{PAR}} = 0.45$, ε is RUE per MJ of intercepted PAR, f_T is the temperature modifier, f_N is the prescribed nutrition factor and f_W is realized transpiration divided by remaining potential transpiration demand after wet-canopy evaporation. The factor 10 converts g m^{-2} to kg ha^{-1} . PAR enters once (Kiniry et al., 1989). The fitted empirical assimilation exponents are 1.85985 for wheat and 0.25000 for maize. They govern carbon response separately from leaf-expansion and leaf-survival sensitivity.

SLA is linearly interpolated through DVS 0, 0.5, 1 and 2, with the DVS-2 value equal to DVS 1. New leaf dry matter acquires area using current SLA; established cohorts retain their birth area-to-mass ratio (de Wit and Berghuijs, n.d.). Thermal senescence hazard $h(\text{DVS})$ is interpolated through DVS 0, 1, 1.4, 1.7 and 2, with a constant preflowering rate and increasing grain-filling rates (Tan et al., 2020). Daily survival

728 is $\exp[-h(\text{DVS}) \max(T_{\text{mean}} - T_{\text{base}}, 0)](1 - q_s)$, with $q_s = 0.025(1 - f_w f_N)$. The terminal hazard
 729 continues at maize R6, whereas wheat BBCH 89 senesces remaining green leaves. LAI emerges from the
 730 cohort area balance; no maximum-LAI parameter or cap enters this formulation.

731 Leaf allocation follows crop development, with separate early and late vegetative multipliers for maize and
 732 additional temperature, water-supply and nutrition limits on leaf expansion. Initial LAI is an emergence condition.
 733 Stage-specific SLA and senescence coefficients retain their inherited estimates and conditional trait priors (APSIM
 734 Initiative, 2026; de Wit, 2022; de Wit, 2020); their values do not represent universal physiological bounds.

735 Grain number is fixed at the first grain-set day as $N_g = 0.1B_f c_g$, where B_f is simulated aboveground dry
 736 mass immediately before grain set and c_g is the grain-number coefficient. Grain set starts at wheat BBCH
 737 61 and maize R1. Maximum grain dry mass is $Q_g = 10N_g m_g$, with fixed m_g of 0.045 g grain⁻¹ for wheat
 738 and 0.300 g grain⁻¹ for maize (APSIM Initiative, 2026). Assimilate supply and development-dependent
 739 filling capacity constrain grain accumulation. Wheat remobilization transfers an explicit subset of stem
 740 mass to grain while conserving total aboveground mass. Live and dead leaves, stems and grain remain in
 741 the aboveground carbon pool; observed yield or harvest index does not initialize grain.

742 Prescribed nutrition factors distinguish archived fertilized and unfertilized management classes. Wuqiao and
 743 regional simulations retain the fertilized class. These factors describe effective growth limitations; the model does
 744 not resolve a mechanistic nitrogen balance.

745 **S2.3. Soil-water balance and actual evapotranspiration**

746 Potential transpiration and exposed-soil evaporation are $T_p = K_T \text{ET}_0 f_c$ and $E_p = K_E \text{ET}_0 (1 - f_c)$, with
 747 ET_0 supplied by AgERA5. Rainfall interception fills a canopy store of capacity 1 mm $\times f_c$. Wet-canopy
 748 evaporation reduces the transpiration demand remaining within each outer water step. Actual ET is
 749 $T_a + E_s + E_c$, the sum of transpiration, soil evaporation and canopy evaporation. Wuqiao residue cover is
 750 unmeasured and its exposed-soil boundary is retained; prescribed organic mulch, where supplied, reduces
 751 potential soil evaporation using the FAO-56 cover approximation (Allen et al., 1998).

752 Two-stage evaporation depends on actual surface-layer storage. Total evaporable water is

753 $\text{TEW} = (\theta_{\text{FC}} - \theta_{\text{AD}})\Delta z_1$ and readily evaporable water is $\text{REW} = 0.30\text{TEW}$. Surface depletion is

754 $D_e = (\theta_{\text{FC}} - \theta_1)\Delta z_1$. The evaporation reduction is $K_r = \min \left[1, \max \left(0, \frac{\text{TEW} - D_e}{\text{TEW} - \text{REW}} \right) \right]$, with actual

755 extraction bounded by storage above air dry. Ponded water evaporates before soil storage. This storage
 756 closure follows the two-stage drying principle (Allen et al., 1998) without resolving a complete dual-crop-
 757 coefficient surface balance.

758 Root activity integrates an exponential depth distribution over rooted parts of each layer and is normalized
 759 over the represented rooted volume. Local depletion stress is $s_i = \min \left[1, \frac{\max(0, W_i - \theta_{WP,i} \Delta z_i)}{(1-p)(\theta_{FC,i} - \theta_{WP,i}) \Delta z_i} \right]$, where p
 760 is the fitted readily available water fraction. Base uptake combines root activity, local stress, atmospheric
 761 demand and the daily extraction coefficient; accessible storage and extraction capacity bound each
 762 contribution. The compensation fraction allocates a bounded part of unmet demand to rooted layers whose
 763 stress factors equal or exceed the activity-weighted mean. Total uptake remains bounded by demand times
 764 the best rooted-layer stress factor and by available extraction capacity. Compensation redistributes
 765 demand and does not eliminate uniformly dry-profile stress.

766 Effective saturation is $S_e = \frac{\theta - \theta_{AD}}{\theta_{SAT} - \theta_{AD}}$. With $m = 1 - \frac{1}{n}$, the retention relation is $S_e = [1 + (\alpha | h |)^n]^{-m}$
 767 (van Genuchten, 1980), and conductivity is $K = K_{sat} S_e^{1/2} [1 - (1 - S_e^{1/m})^m]^2$ (Mualem, 1976; van
 768 Genuchten, 1980). Adjacent half-layer saturated resistances define $K_{sat, face} = \frac{d}{\frac{\Delta z_i}{2K_{sat,i}} + \frac{\Delta z_{i+1}}{2K_{sat,i+1}}}$, where
 769 $d = \frac{\Delta z_i + \Delta z_{i+1}}{2}$. Relative-arithmetic face mobility is $r_{face} = \frac{\Delta z_{i+1} r_i + \Delta z_i r_{i+1}}{\Delta z_i + \Delta z_{i+1}}$, with $r_i = \frac{K_i}{K_{sat,i}}$. Thus

770 $K_{face} = K_{sat, face} r_{face}$. This interface approximation permits a wet layer to supply a dry neighbor while
 771 retaining the resistance of low-conductivity material.

772 Signed interface flux is $q_i = K_{face} \left[1 + \frac{h_i - h_{i+1}}{d} \right]$, positive downward. Transfers have equal and opposite
 773 contributions in neighboring layer balances and are bounded by donor storage and receiving pore capacity.
 774 Twenty-four outer steps partition daily forcing, evaporation and uptake; adaptive internal transport steps
 775 limit net θ increments to 0.005. The free-drainage bottom flux equals the current unsaturated conductivity
 776 under unit hydraulic gradient. Upward redistribution is an internal profile flux, with no external capillary-
 777 rise supply. This bounded explicit layered approximation does not resolve the full implicit Richards
 778 system or saturated positive pressure.

779 For a field of profile storage S , the accounting identity is $\Delta S = P + I - ET_a - D - Q$. Applied irrigation I
 780 is a field-boundary input. Flood-irrigation scenarios wet the entire field, with application, drift and
 781 conveyance losses set to zero at that boundary. Drainage D leaves the modeled profile and is distinct from

782 water reaching an aquifer after deeper vadose-zone transport (Wu et al., 2023).

783 **S2.4. Hydraulic derivation, shared crop parameters and numerical checks**

784 Saxton–Rawls equations use normalized sand and clay fractions and organic matter percent (Saxton and
785 Rawls, 2006). SoilGrids carbon in dg kg^{-1} is converted to organic matter percent by $\frac{1.724 \times \text{SOC}}{100}$; bulk
786 density in cg cm^{-3} is divided by 100; coarse fragments in permille are divided by 1,000 (ISRIC — World
787 Soil Information, 2020). Regional porosity uses $1 - \frac{\text{BD}}{2.65}$, with prescribed density and gravel adjustments.
788 Air-dry water content retains its declared prior. Field-capacity, wilting-point and saturation ordering is
789 checked before simulation; unavailable complete profiles remain excluded under the spatial-source rule.

790 The van Genuchten α and n values are conditional hydraulic priors reconstructed from supplied field-
791 capacity and wilting-point endpoints at pressure heads of $-3,300$ and $-150,000$ mm, with θ_{AD} as residual
792 water and θ_{SAT} as saturated content. Endpoint reconstruction preserves those storage thresholds rather
793 than fitting curves to crop outcomes. Neither depth-resolved Wuqiao retention measurements nor exact
794 seasonal initial layer distributions are available. The separate Wuqiao particle-density prior is 2.75 g cm^{-3} .

795 Table S3 contains effective shared crop values after applying the frozen fit multipliers. Regional and Wuqiao inputs
796 use identical shared cards within each crop. Maximum root depth is limited by the supplied profile depth. Stage-
797 specific SLA and senescence, fixed grain mass and inherited growth coefficients remain distinct from the eight
798 water coefficients fitted in the current run. Near-zero root-density decay yields approximately uniform activity over
799 represented rooted volume; it does not identify an observed root-mass profile.

800 The 164 selected station and field trajectories conserve daily water and aboveground carbon, with maximum
801 absolute residuals of 1.00×10^{-12} mm and 6.71×10^{-12} kg ha $^{-1}$. Exact inputs, observation identities, frozen
802 parameters, source snapshots and SHA-256 provenance accompany the predictions. Conservation establishes
803 arithmetic consistency; hydraulic and crop-response accuracy depend on the conditional observations and
804 retrospective field test.

805 Eight existing water-response coefficients per crop are estimated from documented Wuqiao management in 2016–
806 2018: the transpiration multiplier, soil evaporation coefficient, readily available water fraction, assimilation water-
807 stress exponent, root-density decay, maximum rooting depth, $\log_{10}$ daily root extraction coefficient and root
808 compensation fraction. RUE, leaf allocation and grain-number multipliers remain fixed. Each crop contributes 12
809 seasonal ET, 12 dry-grain and 12 dry-biomass targets, together with three W3–W0 ET contrasts derived from the

810 same seasonal targets. The criterion retains weights of 0.325, 0.175, 0.15 and 0.10 for those components. ET
811 residual scales are $\max(50 \text{ mm}, 0.15 \times \text{observed ET})$; grain and biomass scales are $\max(1,000 \text{ kg ha}^{-1}, 0.20 \times$
812 $\text{observed dry mass})$. Contrast residuals use 50 mm. Eight coefficient-prior terms have total weight 0.01, with
813 inherited bounds, centres and scales.

814 Bounded least squares starts from the inherited water vector, with coefficients scaled to their prescribed intervals, a
815 40-evaluation limit and convergence tolerances of 8×10^{-5} . Reference and refitted coefficients use the identical
816 field-only criterion; the equal-crop mean calibration data loss is 0.592580 for the reference and 0.577934 for the
817 selected refit. Selection excludes prior penalties from this comparison and precedes 2019 testing. Station
818 observations, daily ET and phase ET with unresolved sampling windows have zero weight in the water-parameter
819 fit. Source observations, weather, soil priors, initialization rules and native process equations remain fixed during
820 this refit.

821 **Table S3. Shared crop parameters transferred to regional simulations.**

Parameter	Unit	Wheat	Maize
Radiation-use efficiency	g MJ ⁻¹ intercepted PAR	3.082080	3.379974
Specific leaf area at DVS 0	m ² g ⁻¹	0.028482	0.024045
Specific leaf area at DVS 0.5	m ² g ⁻¹	0.027578	0.029960
Specific leaf area at DVS 1	m ² g ⁻¹	0.023462	0.024000
Leaf senescence rate before flowering	(°C d) ⁻¹	0.000536	0.000010
Leaf senescence rate at DVS 1.4	(°C d) ⁻¹	0.001907	0.000023
Leaf senescence rate at DVS 1.7	(°C d) ⁻¹	0.005327	0.001650
Leaf area index at emergence	m ² m ⁻²	0.141900	0.070844
Grain number per biomass at grain set	grains g ⁻¹	26.272389	4.242740
Maximum individual grain mass	g grain ⁻¹	0.045000	0.300000
Transpiration coefficient	dimensionless	0.838813	1.023273
Soil evaporation coefficient	dimensionless	1.295161	0.100000
Root water uptake coefficient	d ⁻¹	0.042671	0.033543
Leaf biomass partitioning multiplier	dimensionless	1.010797	—
Early vegetative leaf partitioning multiplier	dimensionless	—	0.912989
Late vegetative leaf partitioning multiplier	dimensionless	—	1.779539
Stem reserve fraction	dimensionless	0.270612	—
Stem reserve release fraction	dimensionless	0.067334	—
Assimilation water-stress exponent	dimensionless	1.859851	0.250000
Readily available water fraction	dimensionless	0.543149	0.674994
Root density decay	m ⁻¹	3.41 × 10 ⁻¹⁰	2.27 × 10 ⁻¹³
Maximum rooting depth	mm	1888.654910	1699.999998
Root compensation fraction	dimensionless	1.000000	6.62 × 10 ⁻⁸
Readily evaporable water fraction	dimensionless	0.300000	0.300000

822 Effective shared crop parameters from the selected current cards. RUE uses intercepted PAR with PAR fraction
823 0.45. DVS-2 SLA equals its DVS-1 value; senescence is a thermal hazard with a separate daily stress loss. Wheat
824 BBCH 89 terminates remaining green leaf area; maize retains its terminal hazard at R6. Grain mass is fixed. Water

825 coefficients are estimated from Wuqiao 2016–2018; growth coefficients retain inherited multisite estimates. Root
826 depth is limited by profile depth. Dashes denote crop-specific processes. No maximum-LAI parameter is fitted.

827 **S3. Study-area and spatial aggregation checks**

828 The SRTM-derived terrain grid (NASA JPL, 2013) contains 720 eligible lowland components; the connected
829 component retained by the operational definition covers 404,658 km². Disconnected eligible terrain totals 6,465 km²
830 and contains approximately 7,068 ha of CPM-corrected rotation area. The main definition reaches the southern and
831 eastern bounding-box edges, so geographic sensitivity remains relevant to regional interpretation. Nine elevation–
832 slope alternatives combine elevations 100, 200 and 300 m with slopes 1°, 2° and 3°. Seven annual mapped-area
833 layers cover 2015–2021.

834 A common-extent comparison between ChinaCP plus CPM (Qiu et al., 2022) and the 2020 China Crop Dataset
835 layer (Fu et al., 2026; Fu et al., 2025) yields 9.694 and 7.841 million ha, respectively; CCD2020 is 19.11% lower on
836 that common support. This disagreement is map uncertainty rather than proof of either product’s accuracy. The
837 modeled 2020 area is held fixed to isolate climate and irrigation effects. Historical crop expansion, rotation
838 switching and double-counting of harvested crop area are not estimated.

839 Thirty-two actual-cell representatives use rotation-area weights and selection-period climate means. Eight
840 purposively selected non-representative cells test approximation under all five continuous irrigation fractions. Each
841 crop–fraction comparison retains 96 evaluation cell-years in 2014–2025. Absolute grain, ET and management-
842 response differences compare actual-cell with representative-cell model predictions. The sample measures
843 aggregation sensitivity rather than observed regional accuracy or a probabilistic error bound (Table S4; Figure S1).

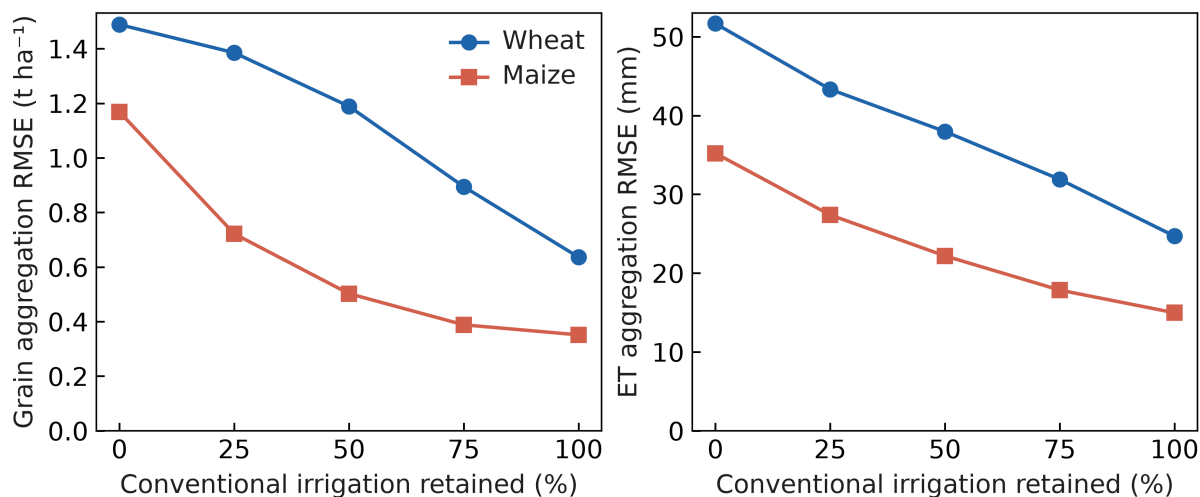
844 **Table S4. Crop-model spatial aggregation errors in eight purposively selected simulation cells.**

Crop	Irrigation fraction	Variable	Unit	N	RMSE	Bias
Maize	0.000	Grain yield	t ha ⁻¹	96	1.17	0.18
Maize	0.000	ET	mm	96	35.21	6.41
Maize	0.250	Grain yield	t ha ⁻¹	96	0.72	0.13
Maize	0.250	ET	mm	96	27.39	5.17
Maize	0.500	Grain yield	t ha ⁻¹	96	0.50	0.08
Maize	0.500	ET	mm	96	22.19	3.82
Maize	0.750	Grain yield	t ha ⁻¹	96	0.39	0.08
Maize	0.750	ET	mm	96	17.85	4.38
Maize	1.000	Grain yield	t ha ⁻¹	96	0.35	0.09
Maize	1.000	ET	mm	96	14.97	5.69
Wheat	0.000	Grain yield	t ha ⁻¹	96	1.49	0.19
Wheat	0.000	ET	mm	96	51.66	-3.18
Wheat	0.250	Grain yield	t ha ⁻¹	96	1.39	0.07
Wheat	0.250	ET	mm	96	43.32	-4.26
Wheat	0.500	Grain yield	t ha ⁻¹	96	1.19	-7.41 × 10 ⁻³
Wheat	0.500	ET	mm	96	37.97	-4.56
Wheat	0.750	Grain yield	t ha ⁻¹	96	0.89	-0.01
Wheat	0.750	ET	mm	96	31.90	-2.05
Wheat	1.000	Grain yield	t ha ⁻¹	96	0.64	0.11
Wheat	1.000	ET	mm	96	24.70	3.21

845 Current model aggregation discrepancies against eight actual-cell simulations under all five irrigation fractions.

846 Grain is converted from kg ha⁻¹ to t ha⁻¹ in every displayed grain-valued column. Exactly zero irrigation

847 discrepancies are omitted from display; full inputs and metrics remain in the current-results product.



848 **Figure S1. Current crop-model spatial aggregation errors under all five irrigation fractions in eight purposively selected cells during**
849 **2014–2025. Actual-cell and representative predictions share the current native model and crop parameters. These discrepancies**
850 **assess spatial representation rather than field-observation accuracy.**

851 **S4. Annual field yields and irrigation-treatment contrasts**

852 Annual Wujiao grain targets comprise sixteen treatment-year means per crop in 2016–2019, digitized with standard
853 errors from the primary Supplementary Figure S2. Grain retains its 13% moisture basis in comparisons. The twelve
854 2016–2018 means per crop enter calibration after a single multiplication by 0.87 to dry matter. The four 2019
855 means remain outside fitting and selection but had been inspected during earlier development. Four-year means
856 summarize the same years and supply no additional independent targets.

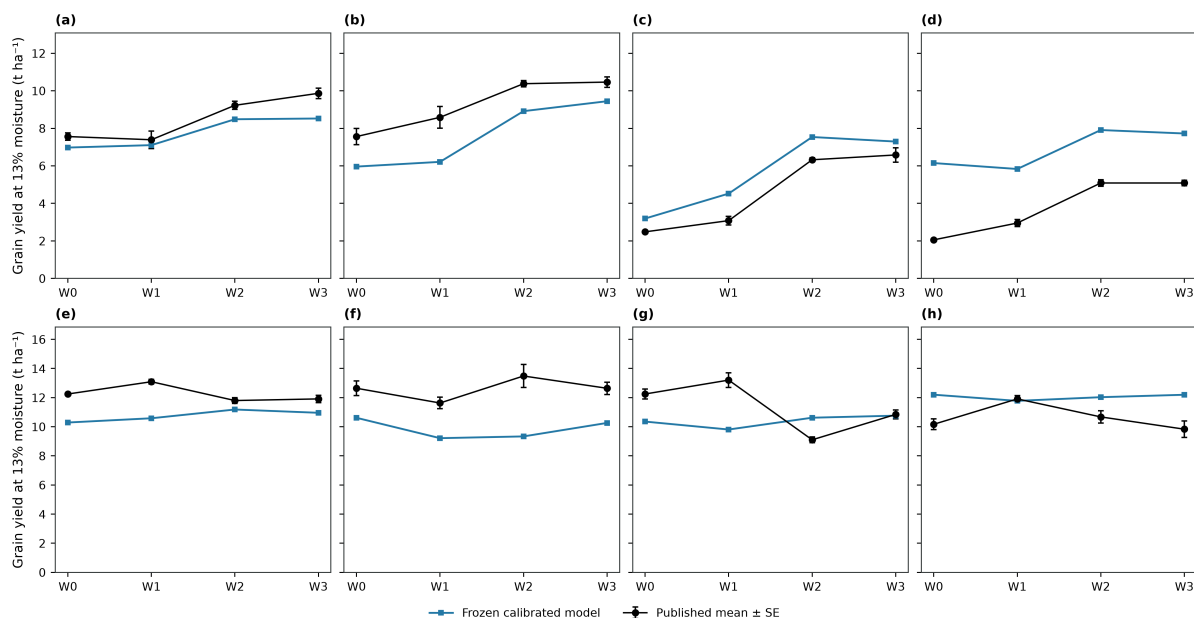
857 Annual grain calibration RMSE is 1.249 t ha⁻¹ for wheat and 2.267 t ha⁻¹ for maize; retrospective-testing RMSE is
858 3.161 and 1.700 t ha⁻¹ at 13% moisture. Within-year W1–W3 minus W0 grain-contrast RMSE is 0.519 and 1.669 t
859 ha⁻¹ in calibration and 1.316 and 1.323 t ha⁻¹ in 2019 (Table S5; Figure S2). Maize treatment labels retain
860 preceding wheat management because maize irrigation is identical within each year.

861 All printed seasonal ET targets remain eligible after matching the 2016 maize rainfall period. Maize-season rainfall
862 of 499.1 mm excludes the 100.5 mm during wheat from the 599.6-mm rotation total. Reconstructed balance
863 residuals for W0–W3 are +0.1, +0.1, -0.9 and +0.1 mm. Source ET and depletion values remain unchanged.
864 Storage-depletion diagnostics share the source ET balance and its unconfirmed exact sampling dates; they are
865 conditional accounting comparisons rather than independent soil-water validation.

866 **Table S5. Wuqiao annual grain and within-year treatment contrasts by fitting and testing partition.**

Crop	Dataset	Comparison	N	RMSE (t/ha)	Bias (t/ha)	NSE	R ²
Maize	Calibration (2016–2018)	Annual grain yield	12	2.267	-1.738	-2.940	0.128
Maize	Calibration (2016–2018)	Within-year W1–W3 minus W0	9	1.669	0.296	-0.788	0.076
Maize	Retrospective evaluation (2019)	Annual grain yield	4	1.700	1.403	-3.641	0.972
Maize	Retrospective evaluation (2019)	Within-year W1–W3 minus W0	3	1.323	-0.836	-1.403	1.000
Wheat	Calibration (2016–2018)	Annual grain yield	12	1.249	-0.445	0.746	0.817
Wheat	Calibration (2016–2018)	Within-year W1–W3 minus W0	9	0.519	0.064	0.857	0.882
Wheat	Retrospective evaluation (2019)	Annual grain yield	4	3.161	3.108	-4.654	0.870
Wheat	Retrospective evaluation (2019)	Within-year W1–W3 minus W0	3	1.316	-1.312	-0.712	0.994

867 Grain is at 13% moisture. Calibration comprises 2016–2018 and retrospective testing comprises 2019. Source
868 means and standard errors are digitized from Supplementary Figure S2, with pixel tolerances near 0.043 t ha⁻¹ for
869 wheat and 0.056 for maize. Contrasts compare W1–W3 with same-year W0. Four-year means overlap these annual
870 targets; maize contrasts retain preceding wheat histories.



871 **Figure S2.** Annual Wuqiao grain at 13% moisture under the selected current parameters. Panels (a–d) show wheat and (e–h) maize;
872 columns from left to right show 2016, 2017, 2018 and 2019. Black points and error bars are published means $\pm$ standard error of
873 three replicates; blue squares show current predictions. The 2016–2018 treatment means contribute to water calibration after
874 conversion to dry grain; 2019 is retrospective testing, excluded from fitting and selection. Maize irrigation is identical within each
875 year and labels retain preceding wheat management. Source means and SE are digitized from Supplementary Figure S2 of the
876 primary experiment.

877 **S5. Class exposure, continuous states and rainfall-only controls**

878 All nine common GRACE-available years occupy lower-storage classes; three other years use conventional
879 fallback. The selected current quotas are identical across storage strata within each rainfall group. Matched rainfall-
880 only and storage–rainfall histories have identical actions, crop states and outcomes throughout the modeled record.
881 Available-rainfall controls apply reduced quotas during monitoring gaps and therefore create different carried water
882 states before later common years. These history effects are separate from the value of the storage classification.
883 Table S7 reports all-year and common-year retention and water use; Table S8 reports the matched and available-
884 rainfall controls.

885 **Table S6. Class exposure and missing-storage coverage by period.**

Period	Class	Unit-years	Years	Area-years (ha yr)	Area-year fraction
evaluation	Lower storage / lower rainfall	174	9	55410278.787	0.468
evaluation	Lower storage / higher rainfall	114	9	33389136.355	0.282
evaluation	Missing storage	96	3	29599805.048	0.250
selection	Higher storage / lower rainfall	68	6	19466918.675	0.179
selection	Higher storage / higher rainfall	124	6	39732691.420	0.366
selection	Lower storage / lower rainfall	109	5	34747673.179	0.320
selection	Lower storage / higher rainfall	51	5	14585335.233	0.134

886 **Table S7. Continuous class-adaptive irrigation evaluation during 2014–2025.**

Strategy	Period	Years	Grain retained (%)	Irrigation cut (mm)	ET cut (mm)	Minimum grain (%)
Storage– rainfall 95%	Nine GRACE years	9	96.270	130.721	36.842	92.115
Storage– rainfall 95%	All twelve years	12	96.818	98.040	29.733	92.115
Storage– rainfall 98%	Nine GRACE years	9	99.626	35.721	6.171	98.579
Storage– rainfall 98%	All twelve years	12	99.622	26.790	5.208	98.579
Conventional	Nine GRACE years	9	100.000	0.000	0.000	100.000
Conventional	All twelve years	12	100.000	0.000	0.000	100.000

887 Current continuous histories. Grain retention is a ratio of period-mean regional dry-grain production to conventional
888 irrigation. ET includes crops and fallows. All-year summaries include conventional fallback; common-year
889 summaries use the nine GRACE-available years.

890 **Table S8. Continuous GRACE–rainfall and rainfall-only strategy comparison during 2014–2025.**

Strategy	Period	Grain retained (%)	Irrigation (mm)	ET (mm)
Storage–rainfall 95%	Nine GRACE years	96.27	249.279	848.527
Storage–rainfall 95%	All twelve years	96.82	281.960	857.698
Storage–rainfall 98%	Nine GRACE years	99.63	344.279	879.197
Storage–rainfall 98%	All twelve years	99.62	353.210	882.223
Rainfall 95%, available	Nine GRACE years	95.40	249.279	844.203
Rainfall 95%, available	All twelve years	95.04	248.565	842.945
Rainfall 95%, matched	Nine GRACE years	96.27	249.279	848.527
Rainfall 95%, matched	All twelve years	96.82	281.960	857.698
Rainfall 98%, available	Nine GRACE years	99.41	344.279	878.151
Rainfall 98%, available	All twelve years	99.36	343.565	879.669
Rainfall 98%, matched	Nine GRACE years	99.63	344.279	879.197
Rainfall 98%, matched	All twelve years	99.62	353.210	882.223

891 Matched controls share the complete GRACE-availability and fallback calendar. Available-rainfall controls act
892 during missing-storage years and alter subsequent soil states. All values use current parameters; differences do not
893 identify causal groundwater responses.



894 **Figure S3. Simulated 0–2 m soil-water storage in representative units (a) 0 and (b) 31. State carries continuously through wheat,**
895 **maize, fallows and annual irrigation choices. Displayed histories compare conventional irrigation with selected current strategies;**
896 **they are model states rather than observed groundwater levels.**

897 **S6. Harvest data quality and crop model recalibration**

898 Original Yucheng harvest sources contain identical six-replicate maize vectors in 2012 and 2017 under different
899 date and cultivar labels. Both groups are retained as raw evidence and excluded from scored grain and harvest
900 biomass. Eligibility depends on source authenticity rather than model residuals. Current plots preserve observed
901 quadrat variation and the original whole site-year partitions (Figure S4).

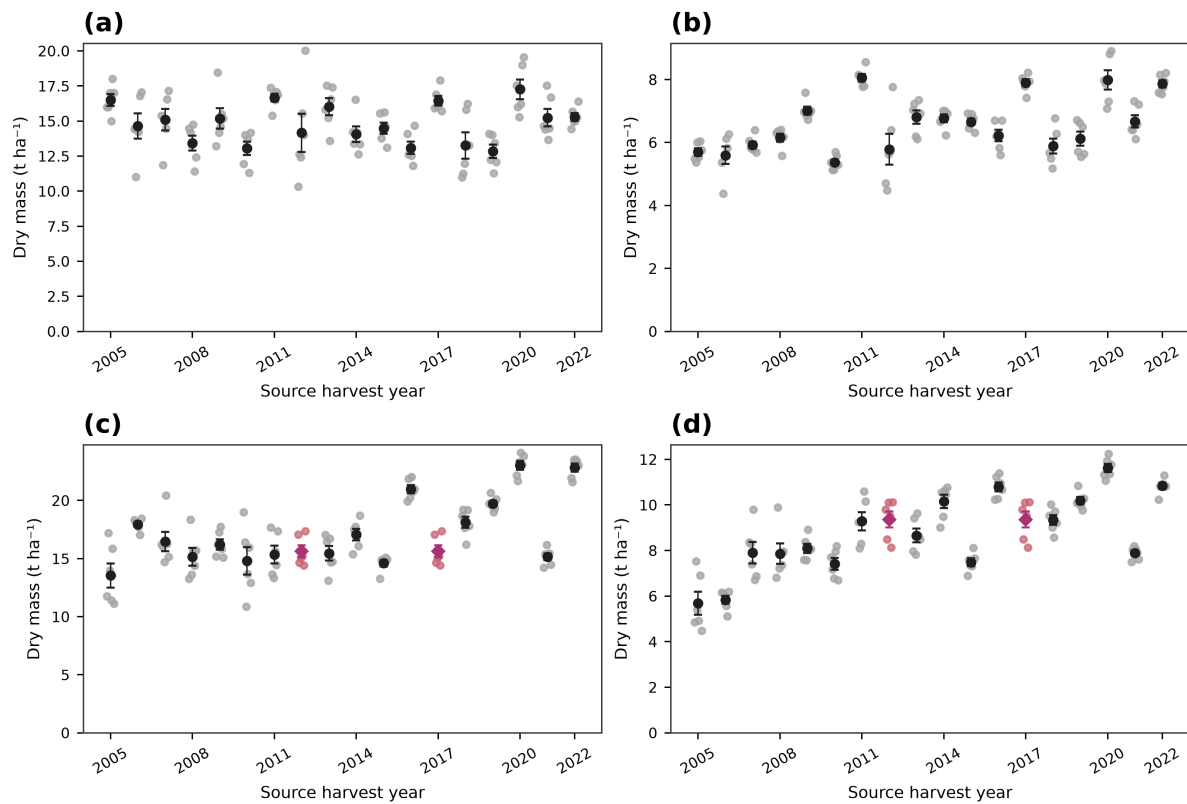


Figure S4. Original Yucheng harvest observations at YCAZH01ABC_01. Panels (a, b) show wheat aboveground biomass and grain; panels (c, d) show maize aboveground biomass and grain. Faint points are six quadrat observations; dark markers and error bars are means $\pm$ replicate standard error. Magenta diamonds identify the identical full maize source vectors in 2012 and 2017. Raw values remain displayed for source inspection; both groups are excluded from comparisons with quality-controlled observations. Error bars describe sampling variation, rather than model or policy uncertainty.

S7. Spatial decomposition under the common irrigation budget

Frozen area shares weight complete current irrigation histories before spatial aggregation. Dry-grain differences in kg ha^{-1} multiply represented hectares and divide by 1,000 for tonnes; water depths multiply hectares by 10 for cubic metres. Independent sums reproduce regional contrasts. Latitude bands describe the distribution of gains and losses, while the 3,641 mapped cells inherit 32 simulated representative responses (Table S9).

912 **Table S9. Spatial decomposition at a common 50% regional irrigation reduction during 2014–2025.**

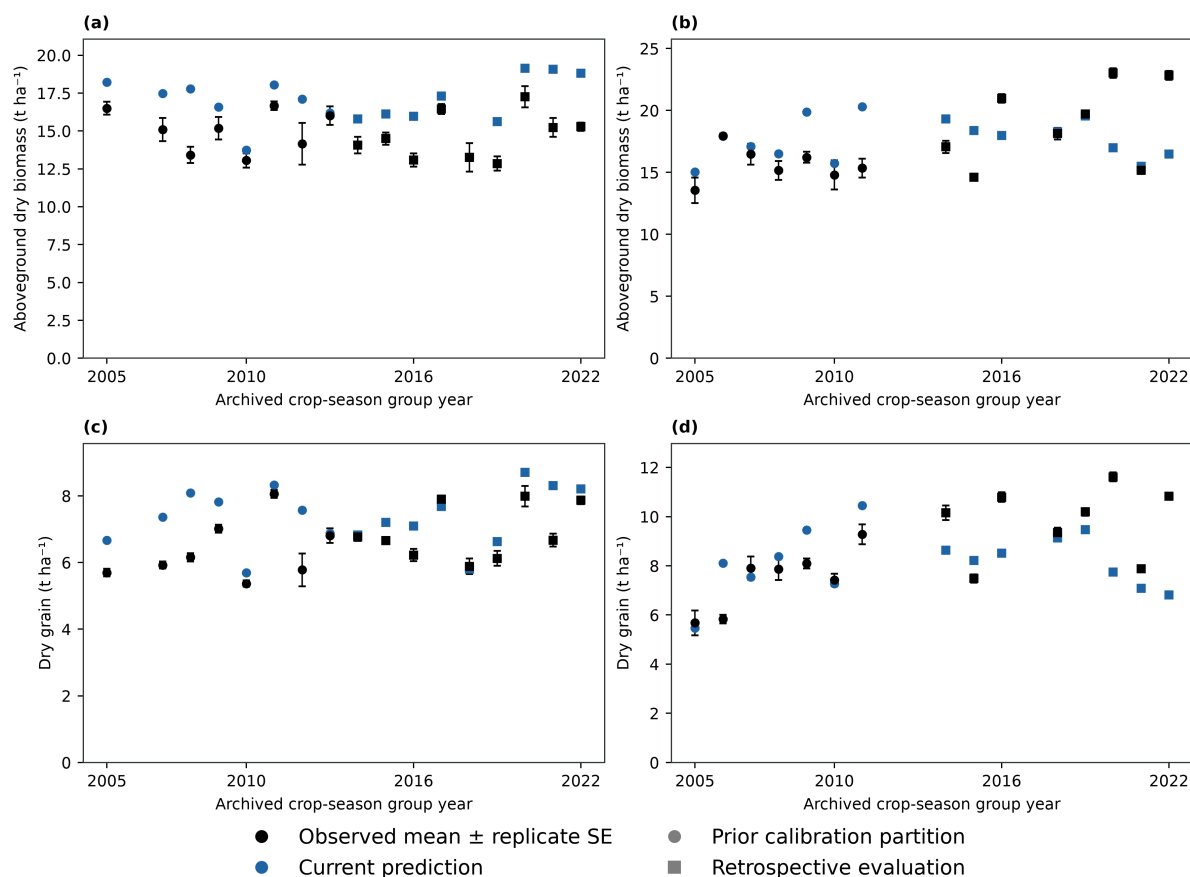
Latitude band	Area share (%)	Targeted irrigation fraction	Grain change (t ha ⁻¹)	ET change (mm)	Grain contribution (Mt)
Southern (32–36°N)	60.733	0.305	-0.357	-25.599	-2.142
Central (36–38°N)	31.369	0.799	2.415	93.497	7.475
Northern (38–41°N)	7.897	0.813	2.420	88.358	1.886

913 Current targeted-minus-uniform means at the same 50% regional irrigation cut. Latitude bands are reporting
914 partitions. Source cells inherit representative responses and are not independent observations.

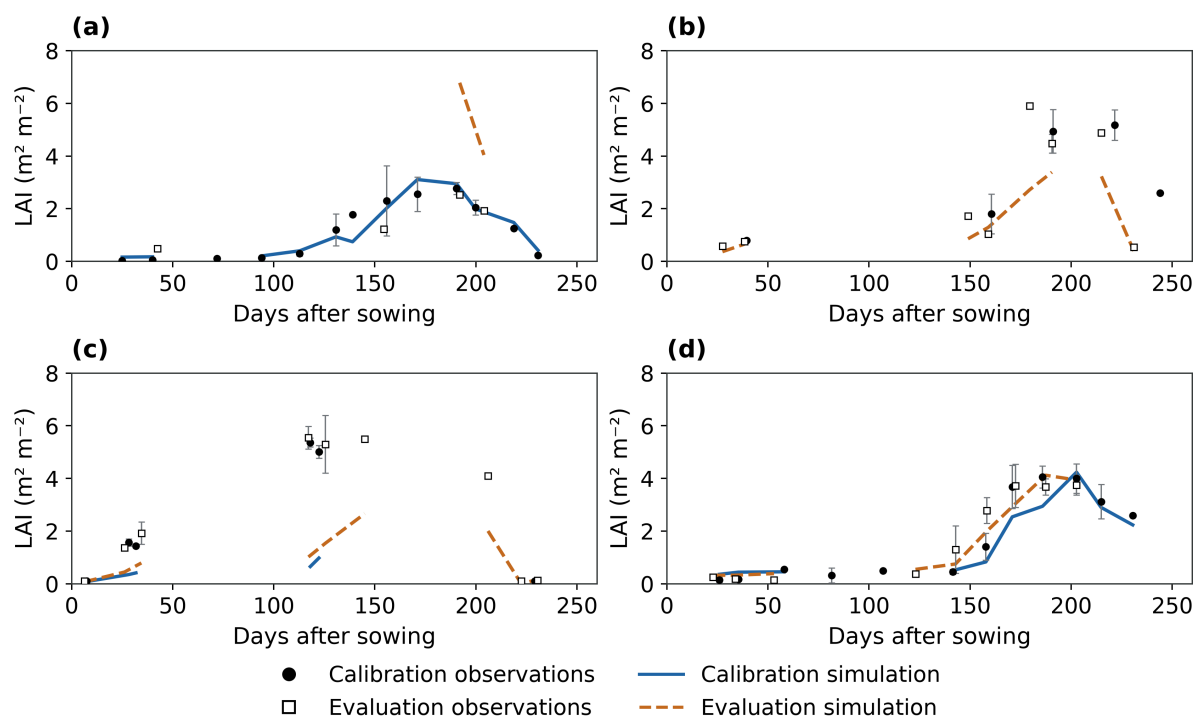
915 **S8. Seasonal crop trajectories and harvest outcomes**

916 Current seasonal curves use the selected frozen parameters and all sites with eligible observations for each variable.
917 Observation and prediction channels share windows, case identities and aggregation weights. Bins span 15 days for
918 wheat and 7 days for maize; site-year contributions and populations can change among bins. Standard errors
919 describe variation across site-year means and are omitted for single-year bins. Missing bins remain gaps. Unbinned
920 records determine performance scores (Figures S6–S11).

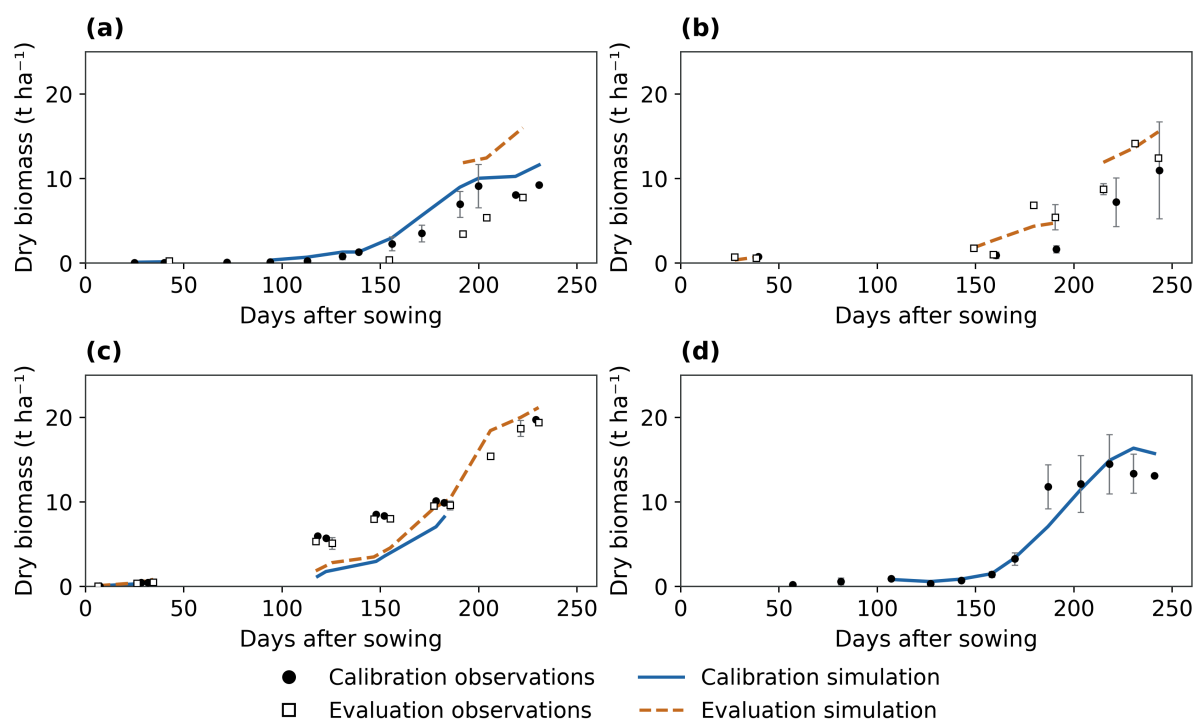
921 The original station calibration partitions retain earlier growth-fitting history; all station comparisons are
922 conditional on unconfirmed complete irrigation management. Station observations have no weight in the latest
923 water fit. Yucheng has no eligible in-season biomass evaluation observations, while its harvest-biomass endpoints
924 remain separately available. Figure S5 shows dry-grain and aboveground harvest means with quadrat SE. Six
925 samples within a crop-year describe sampling variation, and the duplicated 2012 and 2017 maize groups remain
926 excluded. Exact case availability, site-specific metrics, bin contributions and harvest source rows accompany the
927 figures.



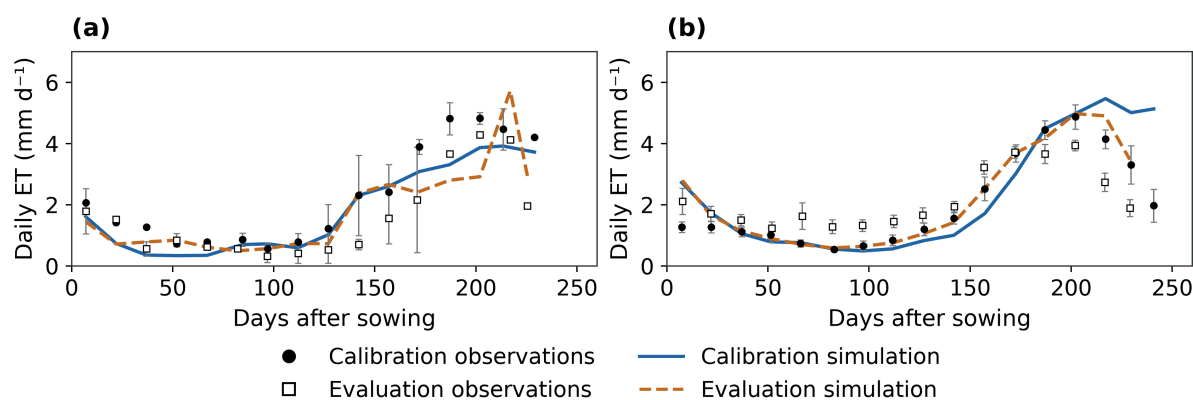
928 **Figure S5. Conditional Yucheng harvest comparisons. Wheat occupies the left column and maize the right; the top row shows**
 929 **aboveground biomass and the bottom row grain. Black points show observed means $\pm$ standard errors among six quadrat samples**
 930 **per crop-year; blue points show current coupled-model predictions at the same sampling windows. Circles and squares distinguish**
 931 **the original calibration and retrospective-evaluation partitions. Grain and total aboveground biomass are dry matter. The duplicated**
 932 **maize 2012 and 2017 source vectors remain excluded. Within-year replicates describe spatial sampling variation and are not**
 933 **independent crop seasons. Management completeness remains unconfirmed and current water coefficients use Wuqiao fitting targets,**
 934 **while growth coefficients retain their earlier multisite fitting history.**



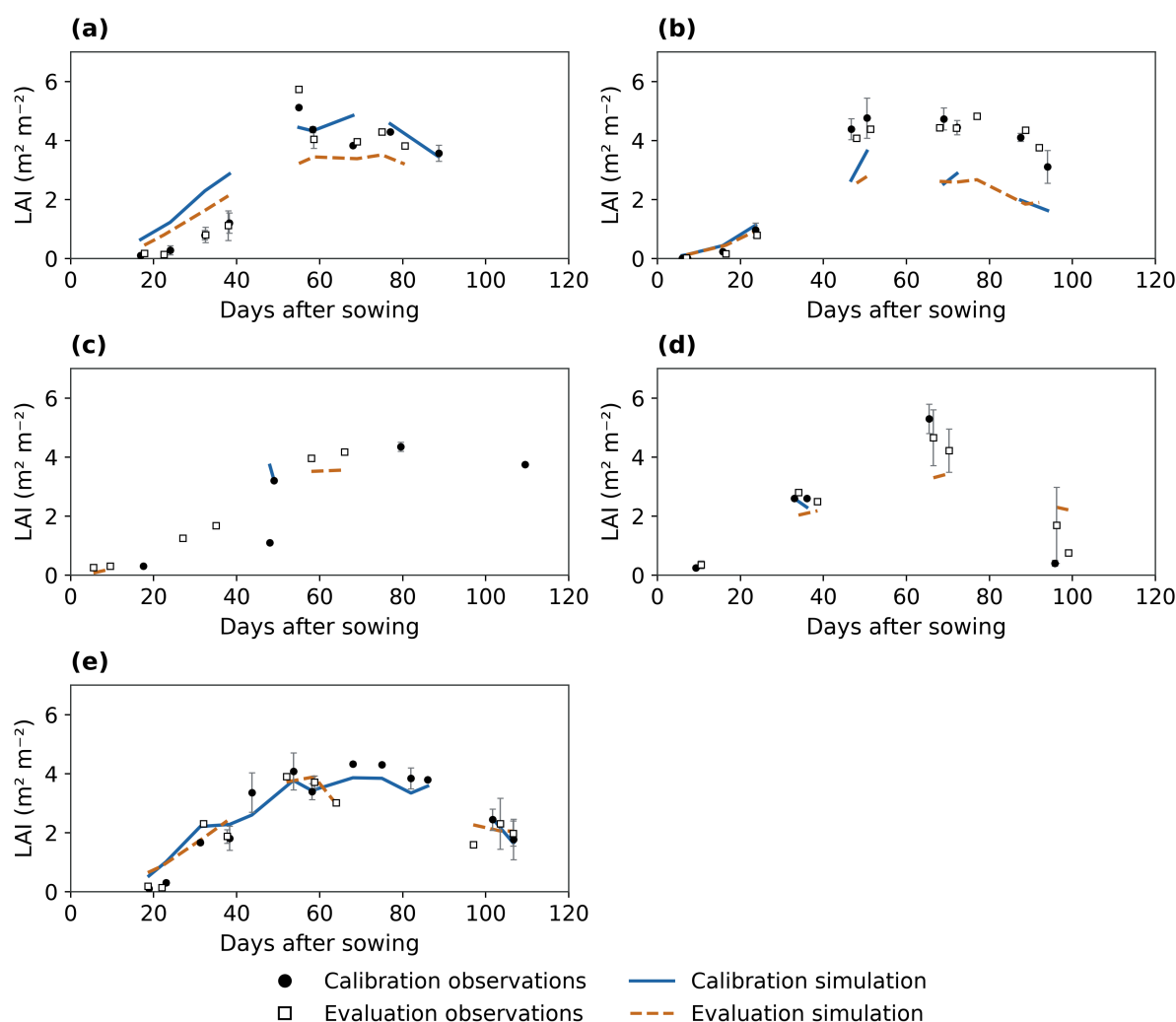
935 **Figure S6. Site-specific wheat LAI comparisons: (a) Fengqiu; (b) Luancheng; (c) Shangqiu; (d) Yucheng. Calibration and**
 936 **retrospective-evaluation observations and simulations share the same axes within each site. Error bars describe site-year**
 937 **errors; gaps are retained. The observations, weighting and conditional management interpretation follow Figure 4.**



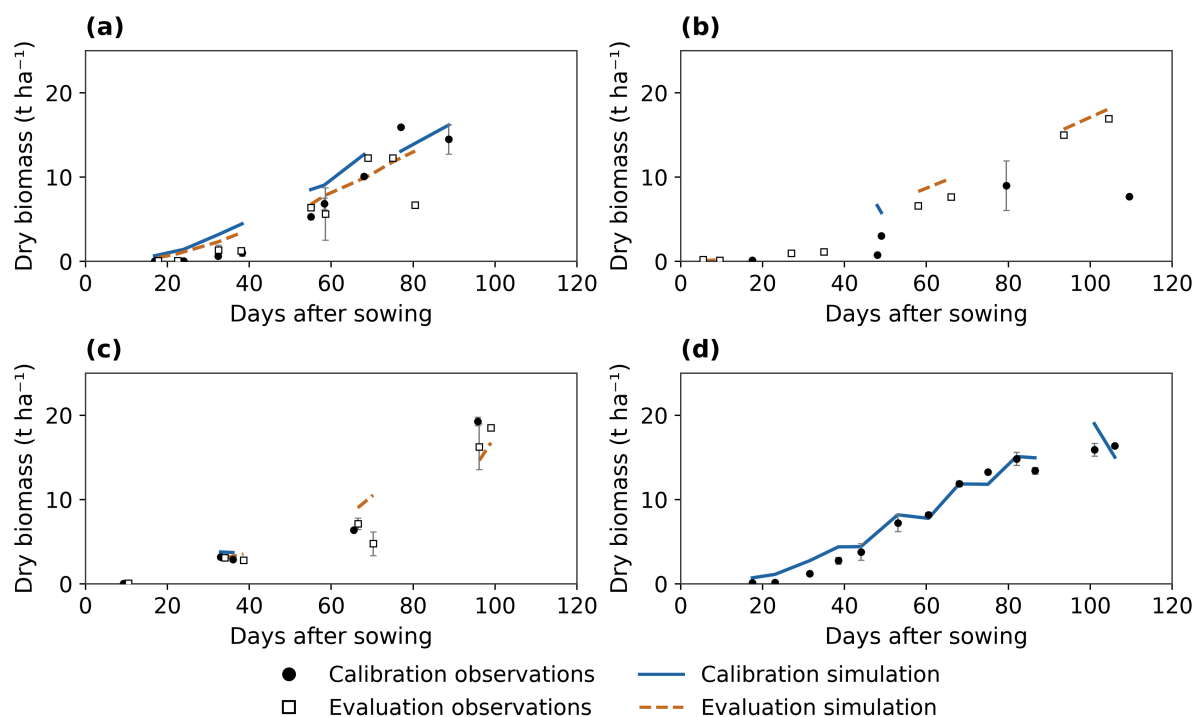
938 **Figure S7. Site-specific wheat aboveground biomass comparisons: (a) Fengqiu; (b) Luancheng; (c) Shangqiu; (d) Yucheng.**
 939 **Calibration and retrospective-evaluation observations and simulations share the same axes within each site. Error bars describe site-**
 940 **year standard errors; gaps are retained. The observations, weighting and conditional management interpretation follow Figure 4.**



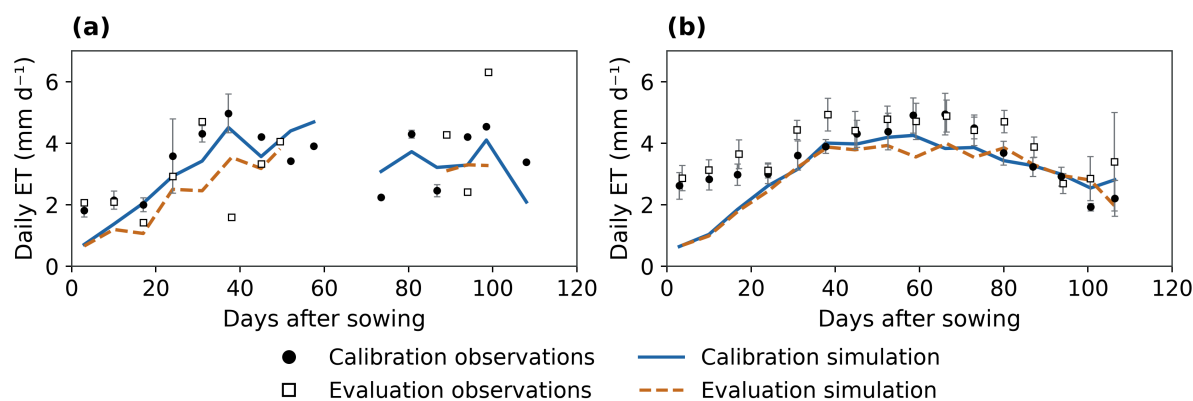
941 **Figure S8. Site-specific wheat ET comparisons: (a) Fengqiu; (b) Yucheng. Calibration and retrospective-evaluation observations and**
 942 **simulations share the same axes within each site. Error bars describe site-year standard errors; gaps are retained. The observations,**
 943 **weighting and conditional management interpretation follow Figure 4.**



944 **Figure S9. Site-specific maize LAI comparisons: (a) Fengqiu; (b) Gucheng; (c) Luancheng; (d) Shangqiu; (e) Yucheng. Calibration**
 945 **and retrospective-evaluation observations and simulations share the same axes within each site. Error bars describe site-year**
 946 **standard errors; gaps are retained. The observations, weighting and conditional management interpretation follow Figure 4.**



947 **Figure S10. Site-specific maize aboveground biomass comparisons: (a) Fengqiu; (b) Luancheng; (c) Shangqiu; (d) Yucheng.**
 948 **Calibration and retrospective-evaluation observations and simulations share the same axes within each site. Error bars describe site-**
 949 **year standard errors; gaps are retained. The observations, weighting and conditional management interpretation follow Figure 4.**



950 **Figure S11. Site-specific maize ET comparisons: (a) Fengqiu; (b) Yucheng. Calibration and retrospective-evaluation observations and**
 951 **simulations share the same axes within each site. Error bars describe site-year standard errors; gaps are retained. The observations,**
 952 **weighting and conditional management interpretation follow Figure 4.**

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